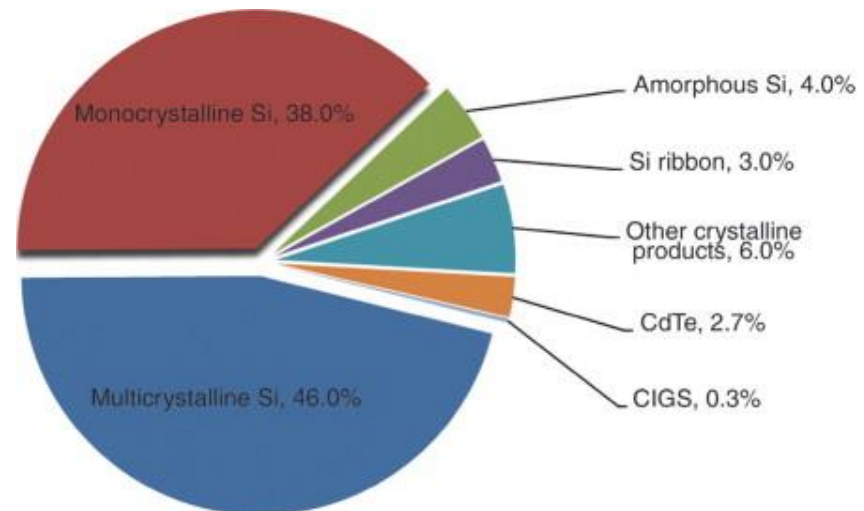


Fundamentals of Photovoltaics (EE336)

Lecture 5 – Crystalline Silicon Solar cells

Crystalline Silicon Solar Cell (c-Si)

- Silicon is not an ideal material for solar cell
 - Bandgap of 1.1eV is not optimum for solar cell (optimum bandgap is 1.4-1.5 eV)
 - Low light absorption (> 300 μm thickness is required to absorb all sun light)
- However, crystalline silicon solar cell is the most popular option for commercial cells.
- More than 80% of the world's photovoltaic market is serviced by crystalline silicon solar cell.



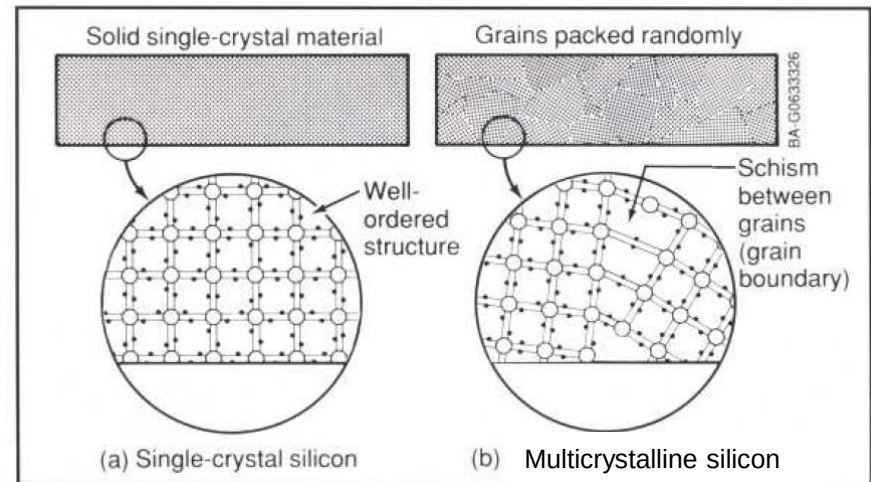
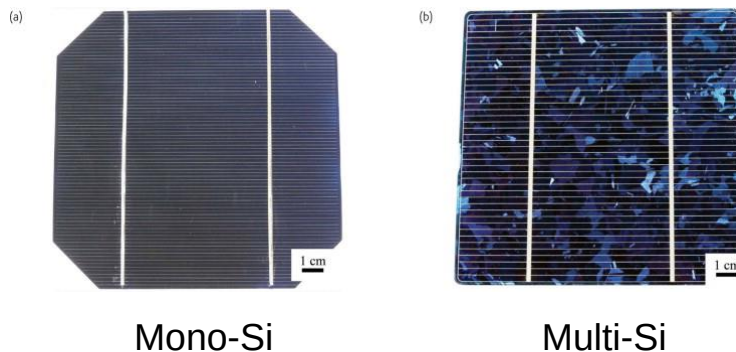
Market share of solar cell types sold during 2006.

Why c-Si solar cell?

- Elemental abundance
 - Second most abundant element in the Earth's crust
- Moderate cost
 - Higher than thin film solar cell but lower than GaAs, InP
- Non-toxicity
- High efficiency (20 – 25 %)
- Device performance stability (lifespan 25 years)
- Physical toughness
- Compatible with the state of art of silicon manufacturing (integrated circuit) technology

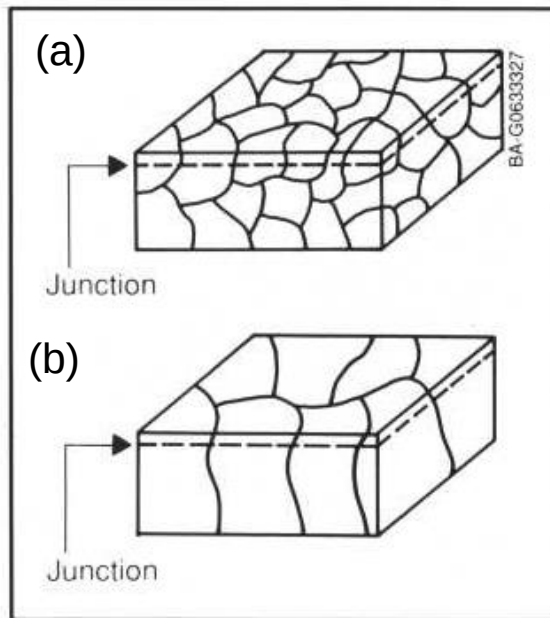
Two types of c-Si solar cell

- Monocrystalline silicon (mono-Si)/ Single crystal silicon
- Multicrystalline silicon (multi-Si)



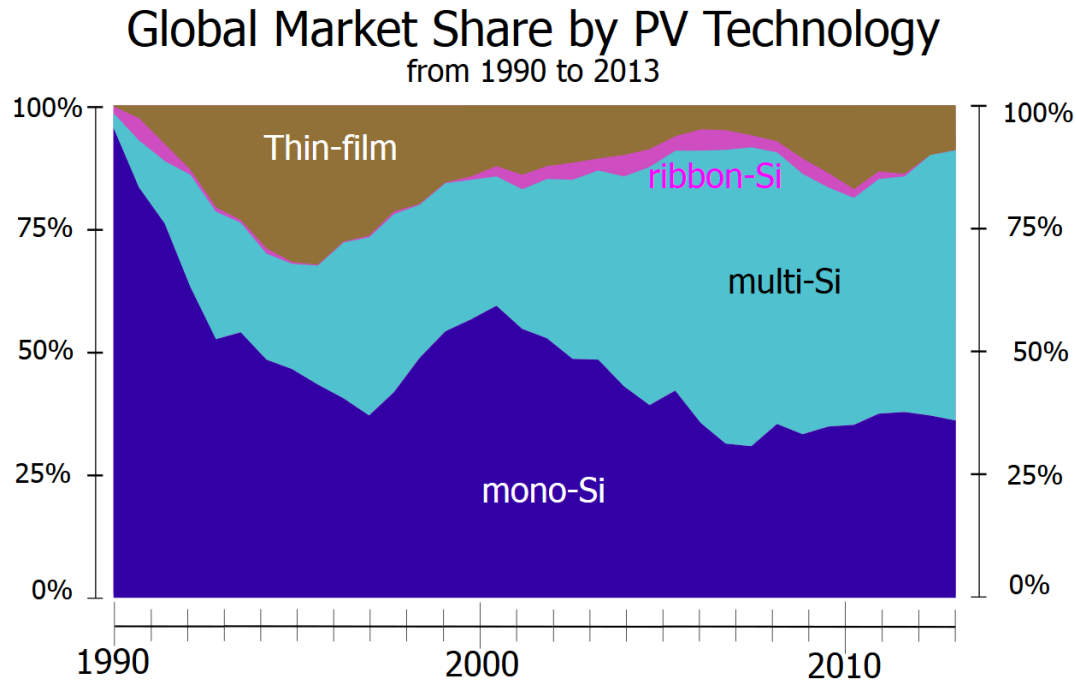
Single-crystal material (a) is structurally uniform; there are no disturbances in the orderly arrangement of atoms. Multicrystalline material (b) is made up of several crystals or grains. At the interfaces of the grains, known as grain boundaries, the atomic order is disrupted. This provides an area where electrons and holes are more likely to recombine.

- Multi-silicon is made up of several crystals, or grains.
- The interface between grains, called “grain boundary”, disrupts the atomic order and provides a place for free electrons and holes to recombine.
- Because of recombination, grain boundaries reduce the power output of a cell.
- Effects of grain boundaries can be minimized through special design, heating and then cooling the material, or filling the broken bonds.

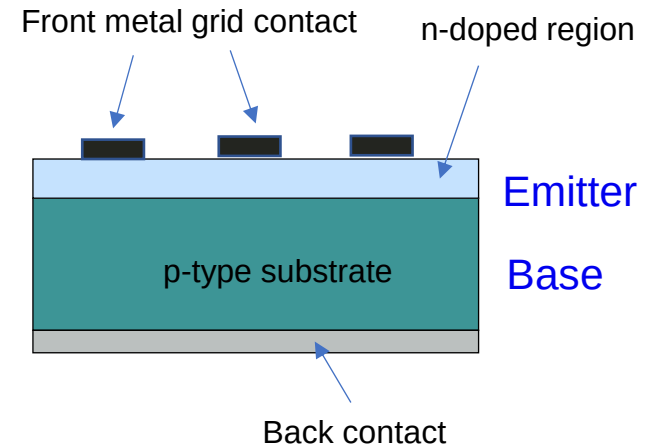
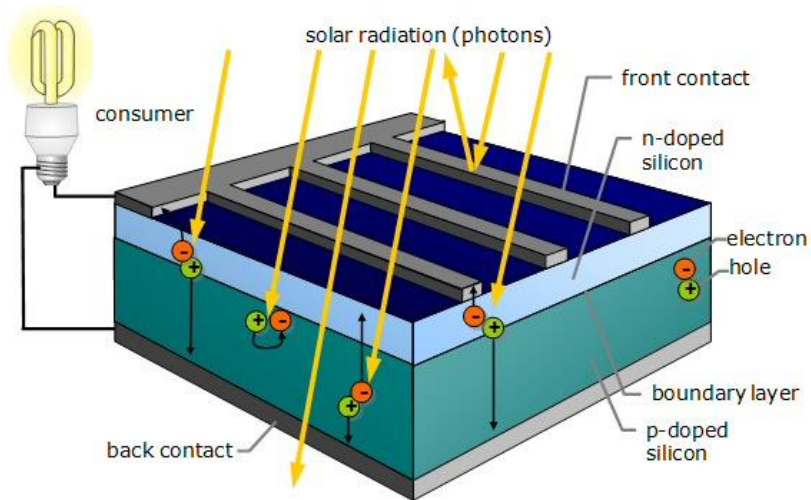


Heating and then cooling Multi-silicon can change randomly oriented grains (a) into larger, columnar ones (b), which alleviates problems with recombination at grain boundaries.

- Multi-Si solar cells are not as efficient as mono-Si solar cells but lower manufacturing costs have made multi-Si very popular.
- In the late 1990s, cells from multi-Si captured almost 33% of the world PV market, compared with about 35% for mono-Si.
- After 2000, multi-Si even takes over the mono-Si in market share.



Design of Silicon Solar Cell



Schematic of a typical c-Si solar cell

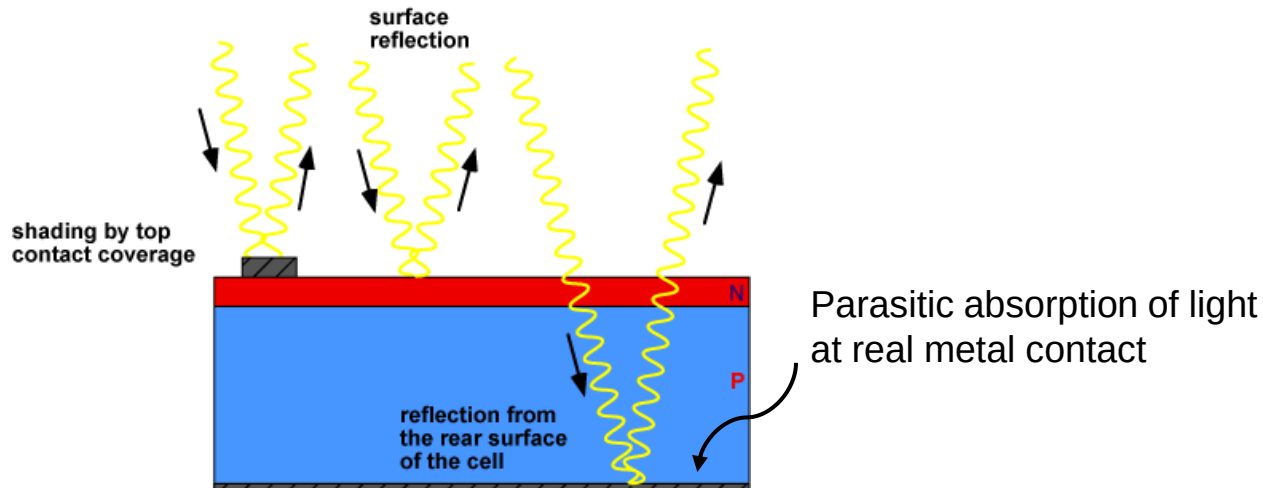
In designing silicon solar cells, the principles for maximizing cell efficiency are:

- increasing the amount of light collected by the cell that is turned into carriers;
- increasing the collection of photo-generated carriers by the p - n junction;
- minimizing the forward bias dark current;
- extracting the current from the cell without resistive losses.

Solar Cell Optics

The principal optical losses in a silicon solar cell are:

- the reflection of light from the front surface (including metal grid contact)
- the parasitic absorption of sunlight within the solar cell at real metal contact
- the reflection at the real metal contact



The reflection of a silicon surface is over 30% due to its high refractive index. The reflectivity, R , between two materials of different refractive indices is determined by:

$$R = \left(\frac{n_0 - n_{Si}}{n_0 + n_{Si}} \right)^2$$

where n_0 is the refractive index of the surrounding and n_{Si} is the refractive index of silicon. n_0 is 1 for an unencapsulated cell and 1.5 for encapsulated cell. The refractive index of silicon changes with wavelength.

There are a number of ways to reduce the optical losses:

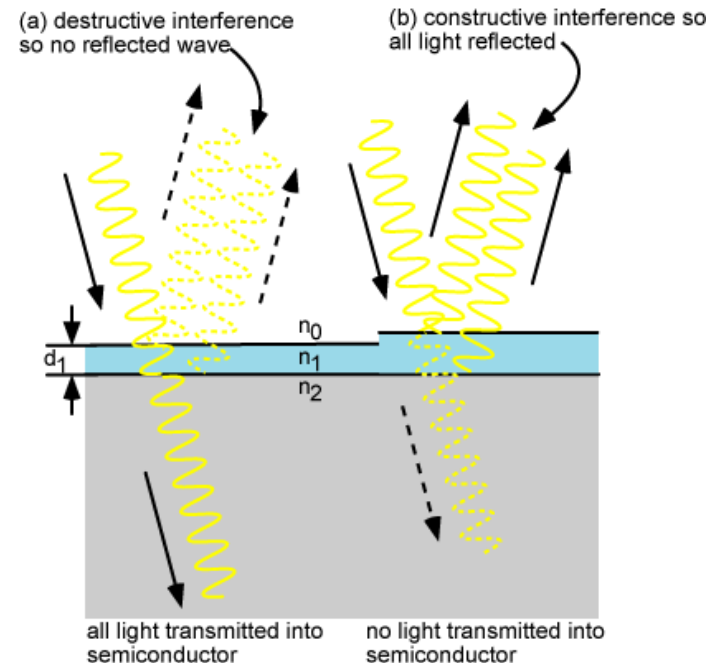
- Top contact coverage of the cell surface can be minimized (although this may result in increased series resistance).
- **Anti-reflection coatings** can be used on the top surface of the cell.
- Reflection can be reduced by **surface texturing**.
- The solar cell can be made thicker to increase absorption (although light that is absorbed more than a diffusion length from the junction has a low collection probability and will not contribute to the short circuit current).
- The **optical path length** in the solar cell may be increased by a combination of surface texturing and light trapping.

Anti-reflection coating (ARC)

The optical losses from the reflection at the front surface can be reduced by anti-reflection coatings. An antireflection coating is a thin transparent film composed of one or several dielectric materials such as [silicon dioxide](#), [silicon nitride](#), [aluminum oxide](#) or [titanium dioxide](#). It works by creating a destructive interference between the incoming and reflected light.

The thickness of the dielectric layer is carefully chosen so that the wave reflected from the top surface of anti-reflection coating to be out of phase with wave reflected from the silicon surface. These out-of-phase reflected waves destructively interfere with one another, resulting in zero net reflective energy.

ARC often is useful for other purposes as well, such as [reducing the recombination](#) of electrons and holes at the silicon surface and providing the [electrical isolation](#) between the silicon and the cell metallization.



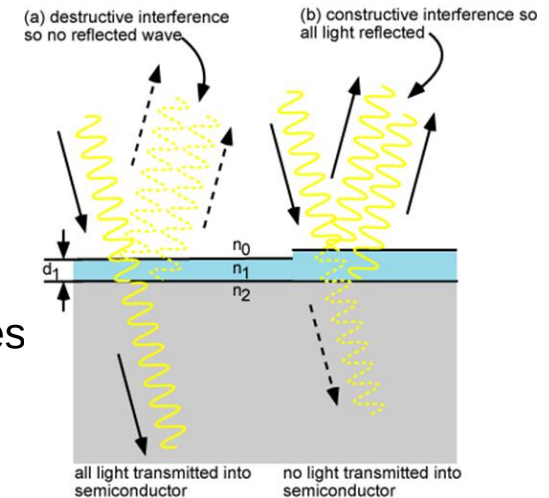
The thickness of the anti-reflection coating is chosen so that the wavelength in the dielectric material is one quarter the wavelength of the incoming wave.

$$d_1 = \frac{\lambda_0}{4n_1}$$

where, n_1 is the refractive index of anti-reflection coating

λ_0 is wavelength of light in free-space

d_1 is the thickness of the dielectric layer which gives minimum reflection



Reflection is further minimized if the refractive index of the anti-reflection coating is the geometric mean of that of the materials on either side; that is, glass or air and the semiconductor. This is expressed by:

$$n_1 = \sqrt{n_0 n_2}$$

For the reflectance at normal incidence we define a series of parameters: r_1 , r_2 , and θ . The surrounding region has a refractive index of n_0 , the ARC has a refractive index of n_1 and a thickness of t_1 , and the silicon has a refractive index of n_2 .

$$r_1 = \frac{n_0 - n_1}{n_0 + n_1}$$

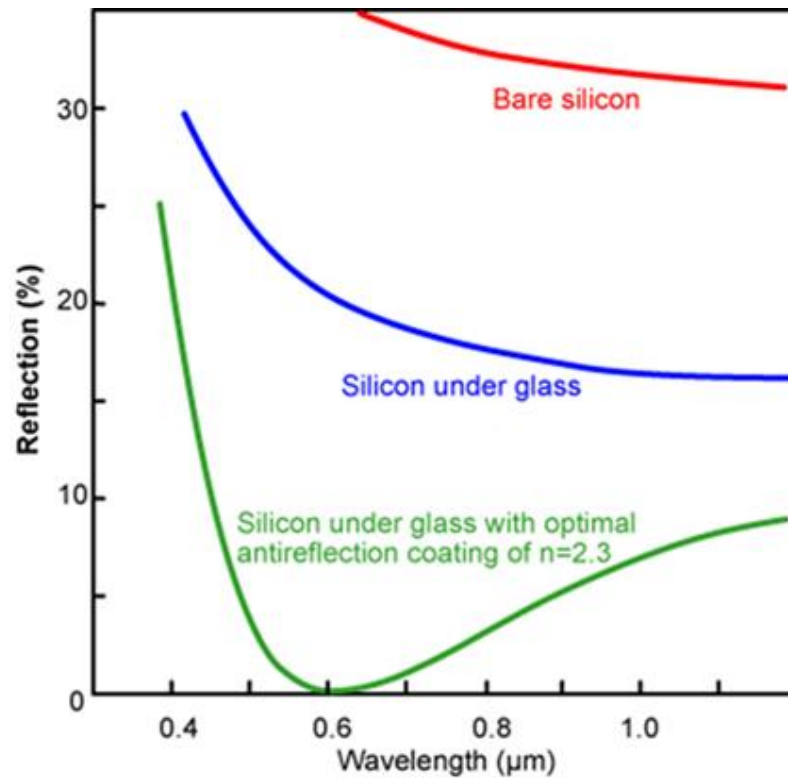
$$r_2 = \frac{n_1 - n_2}{n_1 + n_2}$$

$$\theta = \frac{2\pi n_1 t_1}{\lambda}$$

For a single layer ARC on a substrate the reflectivity is:

$$R = |r^2| = \frac{r_1^2 + r_2^2 + 2r_1 r_2 \cos 2\theta}{1 + r_1^2 r_2^2 + 2r_1 r_2 \cos 2\theta}$$

While the reflection for a given thickness, index of refraction, and wavelength can be reduced to zero using the equations above, the index of refraction is dependent on wavelength and so zero reflection occurs only at a single wavelength. For photovoltaic applications, the refractive index, and thickness are chosen in order to minimize reflection for a wavelength of 0.6 μm . This wavelength is chosen since it is close to the peak power of the solar spectrum.



Comparison of surface reflection from a silicon solar cell, with and without a typical anti-reflection coating.

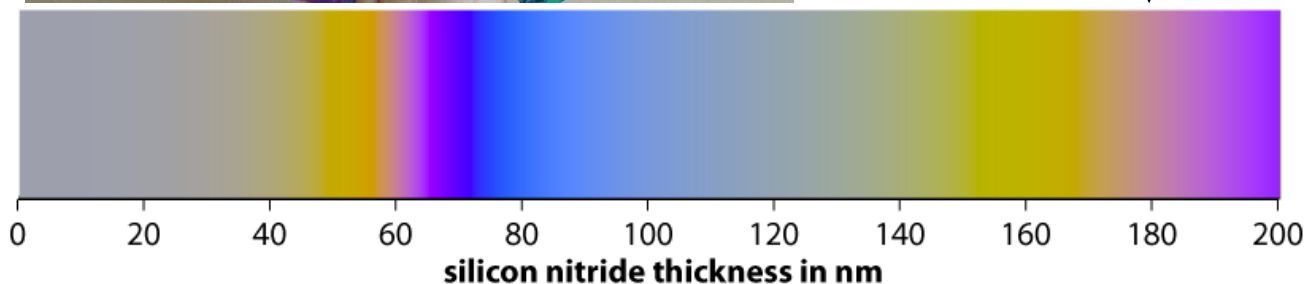
- The most common method to precisely measure the thickness of anti-reflection coatings is using ellipsometry which looks at the way polarized light is reflected.
- A simple method to measure anti-reflection coating is by simply looking at the film colour.
- The colour of the film is affected by the thickness as well as the refractive index so the film color is merely a rough guide to thickness.



Four multicrystalline wafers covered with films of silicon nitride. The difference in colour is solely due to the thickness of the film. The green wafers are very thick films and so don't appear in the color chart



Color of silicon nitride films with a refractive index of ~ 2.05 as a function of film thickness under fluorescent lighting for normal incident light.

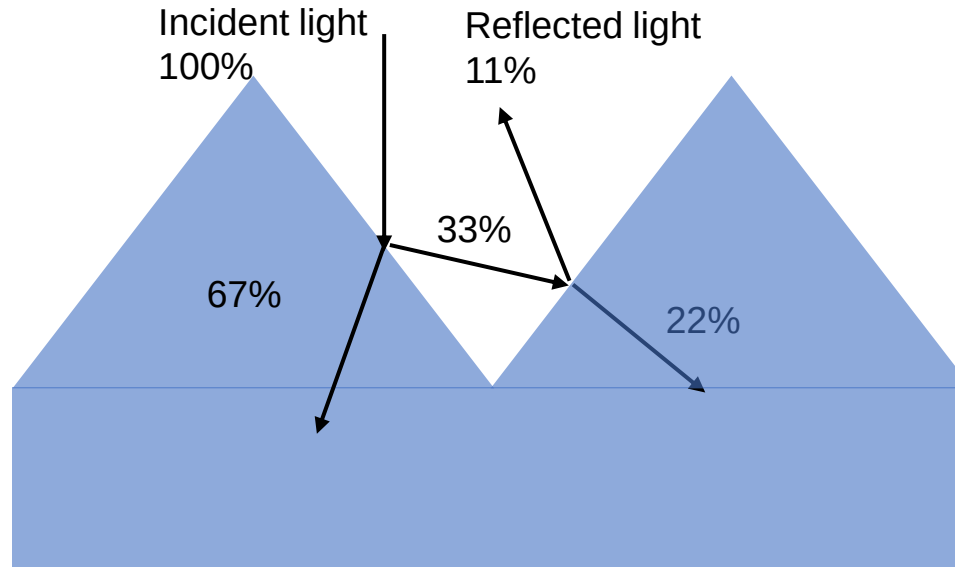


Surface Texturing

Surface texturing, either in combination with an anti-reflection coating or by itself, can also be used to minimize reflection. Any "roughening" of the surface reduces reflection by increasing the chances of reflected light bouncing back onto the surface, rather than out to the surrounding air.

With an ideally textured surface, the amount of incident light that is reflected from the uncoated silicon surface is reduced to approximately 11%.

The combination of texturing and antireflection coating reduces the reflection from the front surface to a few percent.



Surface texturing can be accomplished in a number of ways.

A single crystalline substrate can be textured by etching along the faces of the crystal planes. The crystalline structure of silicon results in a surface made up of pyramids if the surface is appropriately aligned with respect to the internal atoms.

This type of texturing is called "random pyramid" texture, and is commonly used in industry for single crystalline wafers.

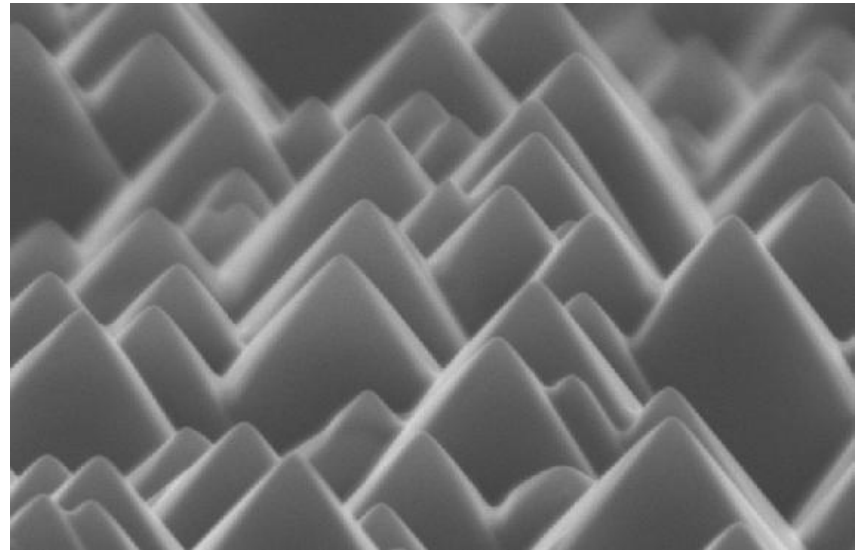
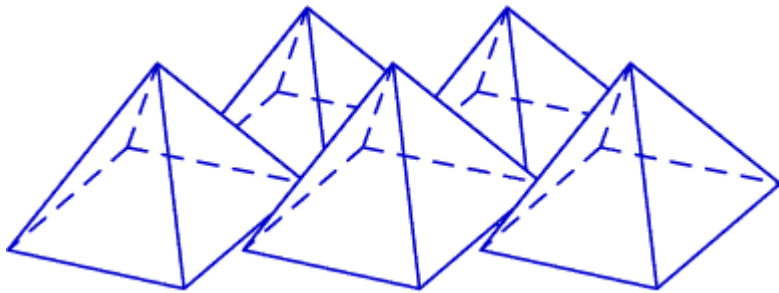
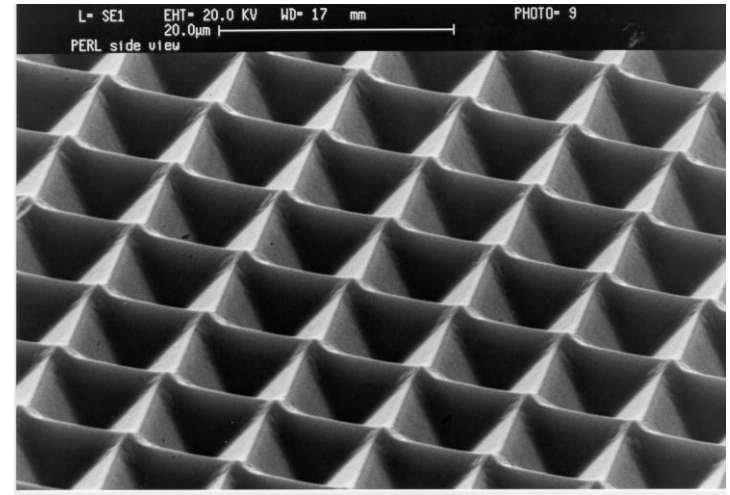


Image Courtesy of The School of Photovoltaic & Renewable Energy Engineering, University of New South Wales.

Another type of surface texturing used is known as "inverted pyramid" texturing. Using this texturing scheme, the pyramids are etched down into the silicon surface rather than etched pointing upwards from the surface.



For multicrystalline wafers, only a small fraction of the surface will have the required orientation of $\langle 100 \rangle$ and consequently these techniques are less effective on multicrystalline wafers. However, multicrystalline wafers can be textured using a photolithographic technique as well as mechanically sculpting the front surface using dicing saws or lasers to cut the surface into an appropriate shape.

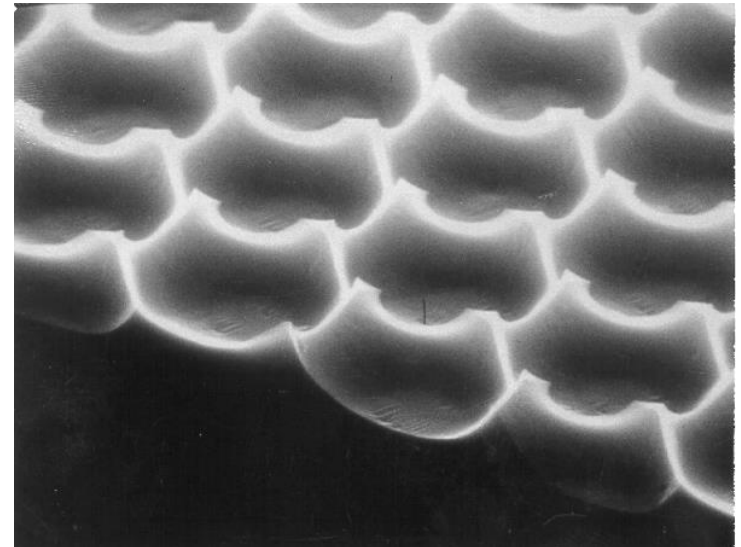
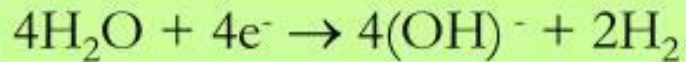
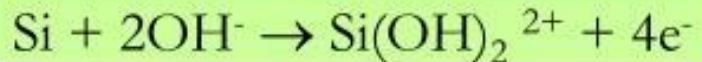


Image Courtesy of The School of Photovoltaic & Renewable Energy Engineering, University of New South Wales.

Anisotropic Etching of Silicon

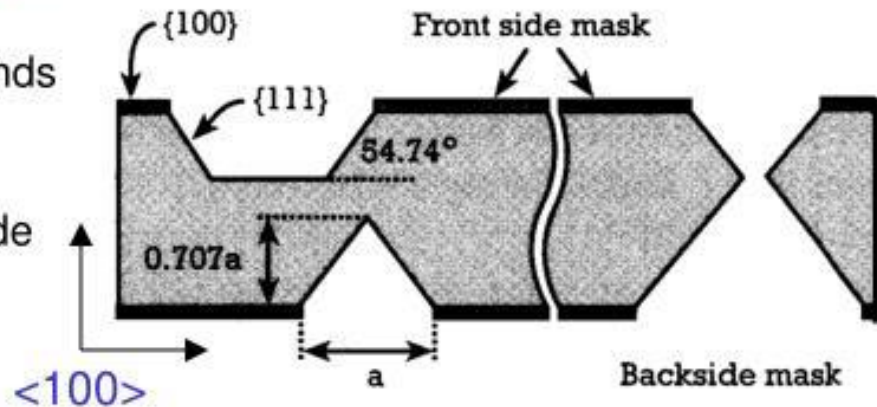
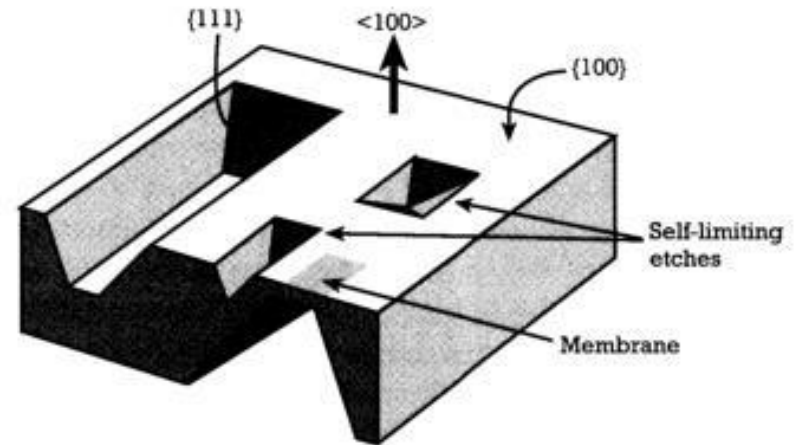
- Etching of Si with KOH



- Crystal orientation relative etch rates

- $\{110\}:\{100\}:\{111\} = 600:400:1$

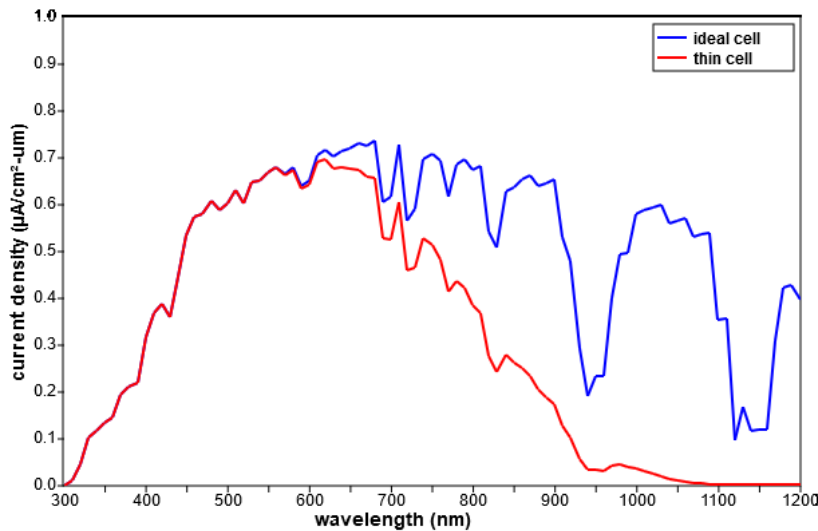
- $\{111\}$ plane has three backbonds below the surface
- Energy explanation
- $\{111\}$ may form protective oxide quickly



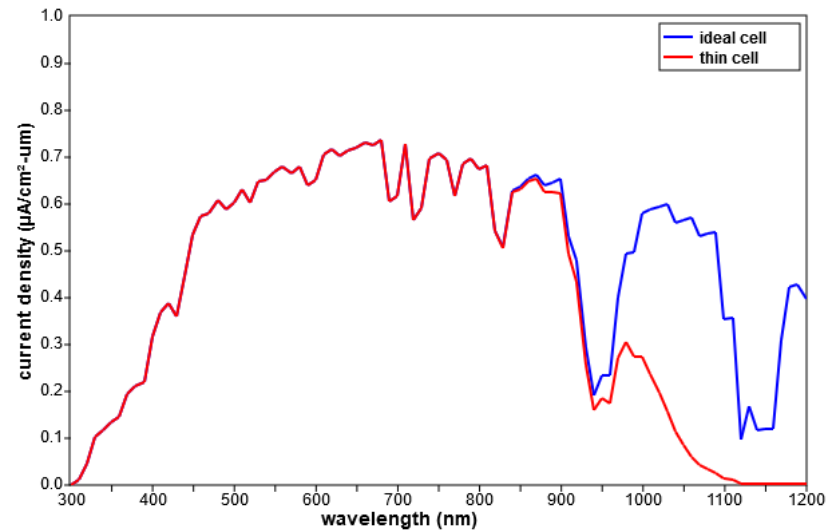
Material thickness

The absorption of light as much as possible is another way to reduce the optical losses. The amount of light absorbed depends on the optical path length and the absorption coefficient.

The below figures show the dependence of photon absorption on device thickness for a silicon solar cell.

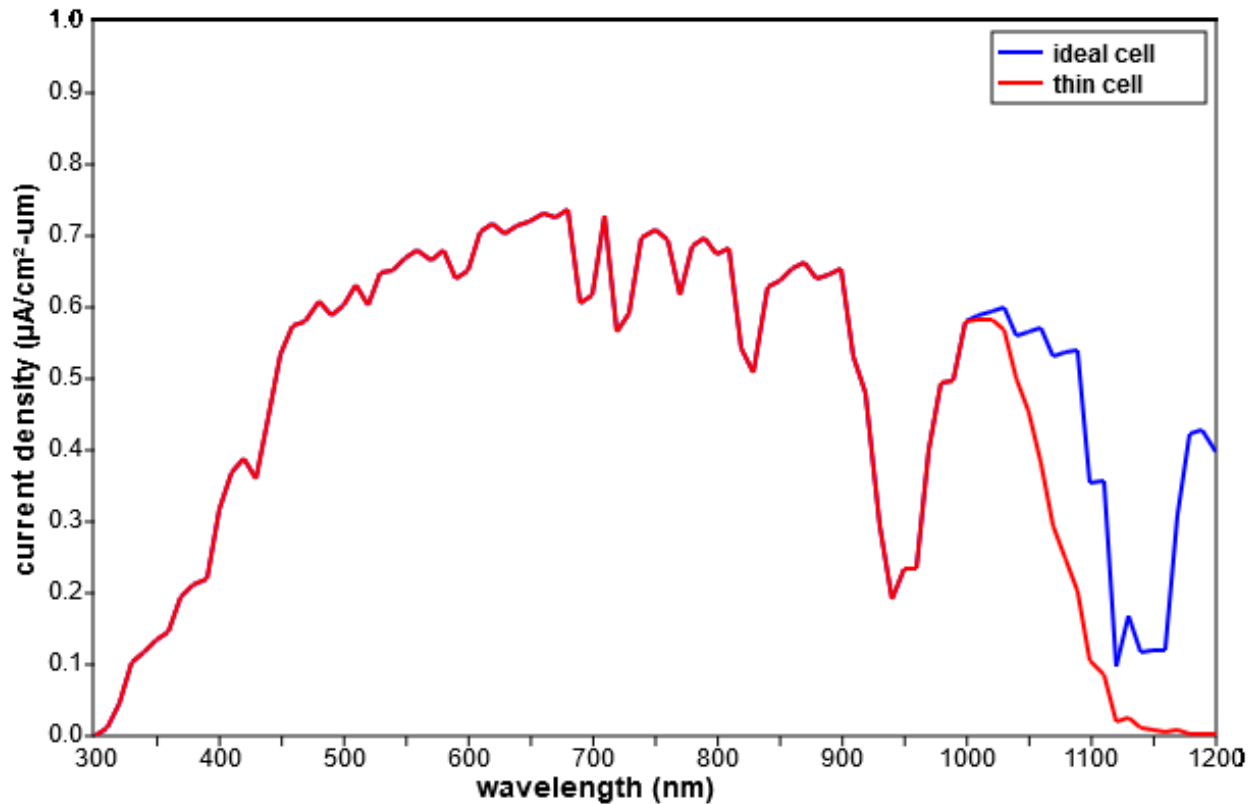


Cell Thickness: 10 microns
Actual J_{sc} : 27.3 mA/cm^2



Cell Thickness: 100 microns
Actual J_{sc} : 36.6 mA/cm^2

Material thickness



Cell Thickness: 1000 microns
Actual J_{sc} : 40.9 mA/cm^2

Light Trapping

The optimum device thickness is not controlled solely by the need to absorb all the light. For example, if the light is not absorbed within a diffusion length of the junction, then the light-generated carriers are lost to recombination.

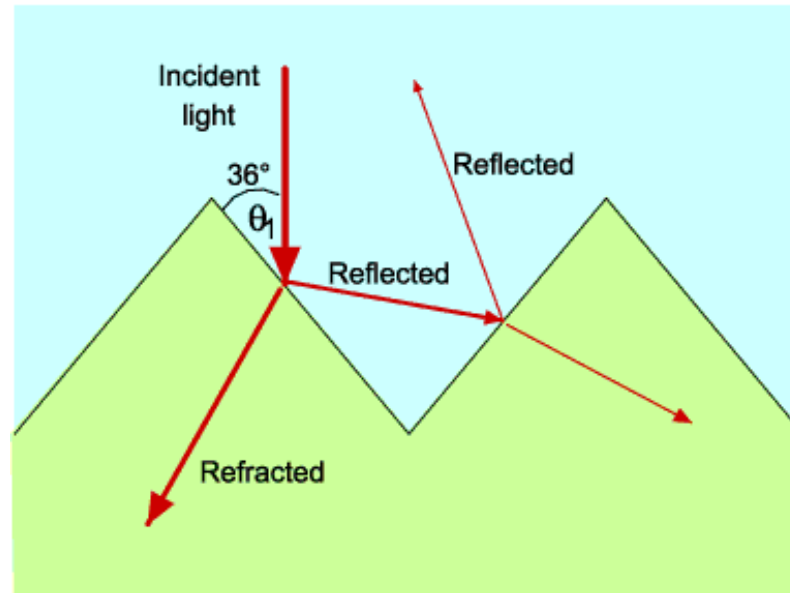
In addition, a thinner solar cell which retains the absorption of the thicker device may have a higher voltage.

Consequently, an optimum solar cell structure will typically have "light trapping" in which the optical path length is several times the actual device thickness. For example, a solar cell with no light trapping features may have an optical path length of one device thickness, while a solar cell with good light trapping may have **an optical path length of 50**, indicating that light bounces back and forth within the cell many times.

(Note: **the optical path length of a device refers to the distance that an unabsorbed photon may travel within the device before it escapes out of the device. This is usually defined in terms of device thickness**)

Light trapping is usually achieved by changing the angle at which light travels in the solar cell by having it be incident on an angled surface. A **textured surface** will not only reduce reflection as previously described but will also couple light obliquely into the silicon, thus giving a longer optical path length than the physical device thickness.

In a textured single crystalline solar cell, the presence of crystallographic planes make the angle θ_1 equal to 36° as shown below.



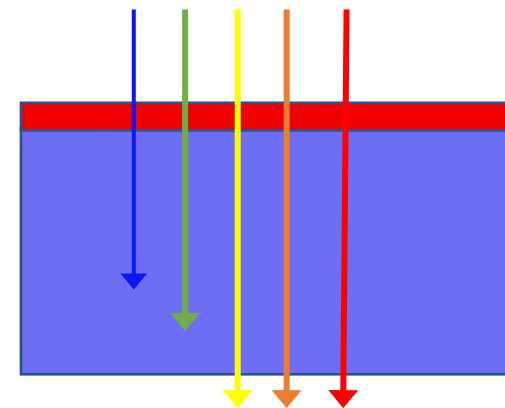
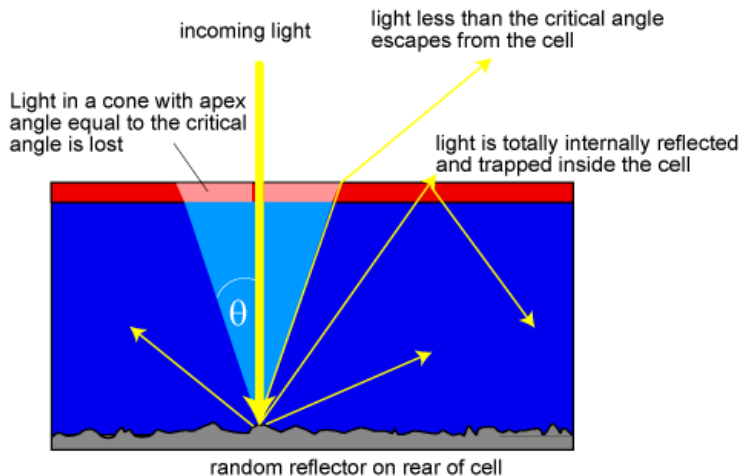
If light passes from a high refractive index medium to a low refractive index medium, there is the total internal reflection when the incident angle is higher than critical angle.

Using total internal reflection, light can be trapped inside the cell and make multiple passes through the cell, thus allowing even a thin solar cell to maintain a long optical path length.

Lambertian Rear Reflectors

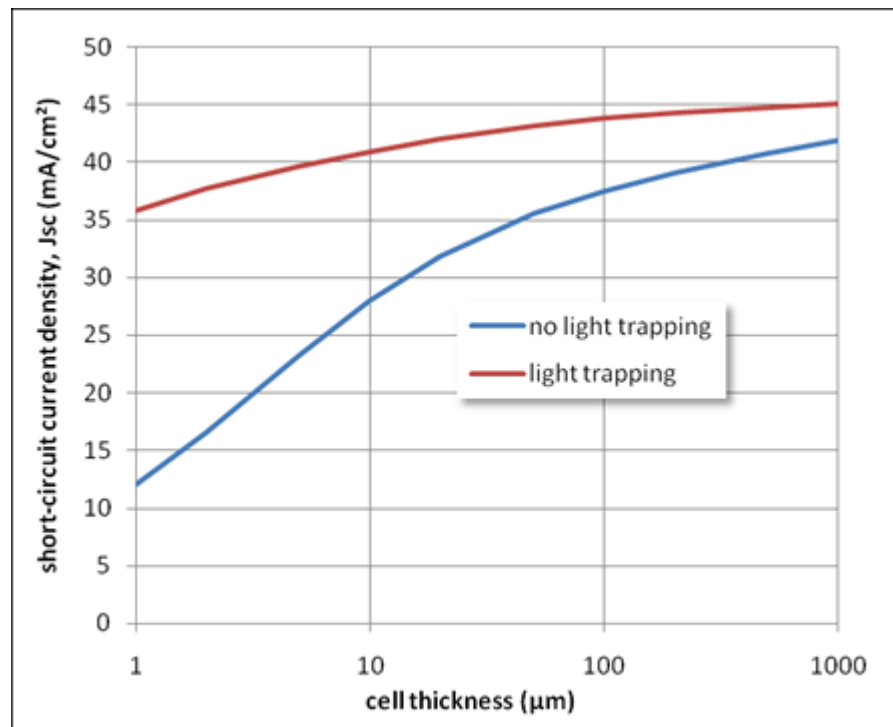
A Lambertian back reflector is a special type of rear reflector which randomizes the direction of the reflected light. **High reflection of the rear cell surface** reduces absorption in the rear cell contacts or transmission from the rear, allowing the light to bounce back into the cell for possible absorption. Randomizing the direction of light allows much of the reflected light to be totally internally reflected.

Light reaching the top surface at an angle greater than the critical angle for total internal reflection is reflected again towards the back surface. Light absorption can be dramatically increased in this way, since **the pathlength of the incident light can be enhanced by a factor up to $4n^2$** where n is the index of refraction for the semiconductor. This allows an optical path length of approximately 50 times the physical devices thickness and thus is an effective light trapping scheme.



Cell without reflector

The graph below shows the short-circuit current of a silicon solar cell with and without light trapping. In the case without light trapping, the light makes only one pass through the solar cell with some of the light (particularly in the infrared) passing straight through the cell without being absorbed. In the case with ideal light trapping the path length is increased by $4n^2$ ($4 \times 3.5^2 \approx 50$) and the short-circuit current is much higher. In practice the level of light trapping is somewhere between these two limits.

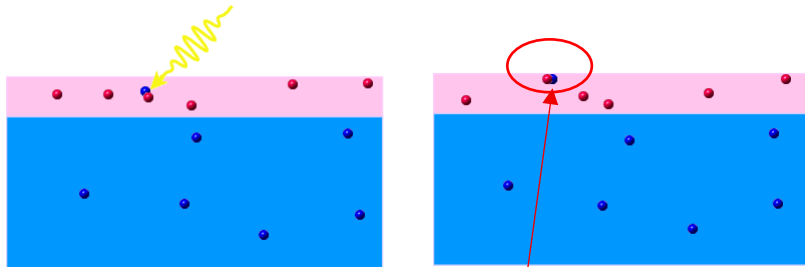


Recombination losses

Recombination losses effect both the current collection (and therefore the short-circuit current) as well as the forward bias injection current (and therefore the open-circuit voltage). Recombination is frequently classified according to the region of the cell in which it occurs. Typically, recombination at the surface (**surface recombination**) or in the bulk of the solar cell (**bulk recombination**) are the main areas of recombination. The depletion region is another area in which recombination can occur (**depletion region recombination**).

Scenario 1

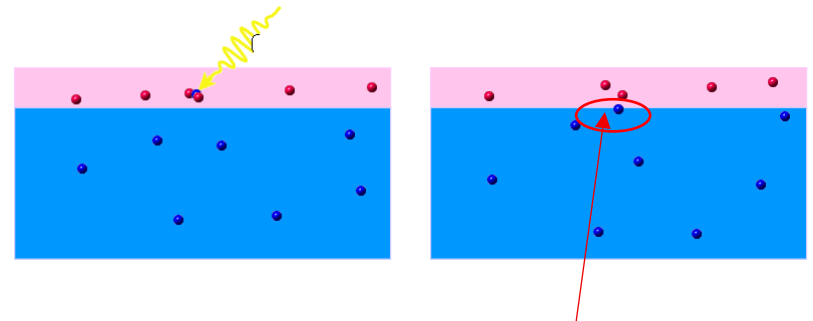
Photon generates an electron-hole pair near the front surface



Minority carrier (hole) recombines at the front surface

Scenario 2

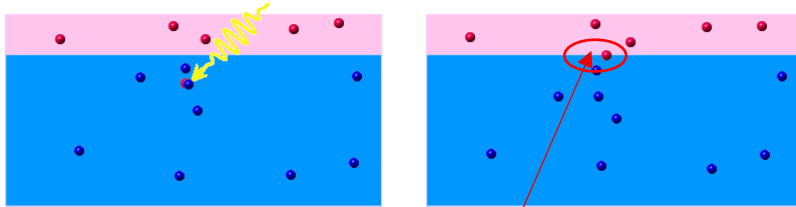
Photon generates an electron-hole pair deeper in the emitter



Minority carrier (hole) is collected when it crosses the junction

Scenario 3

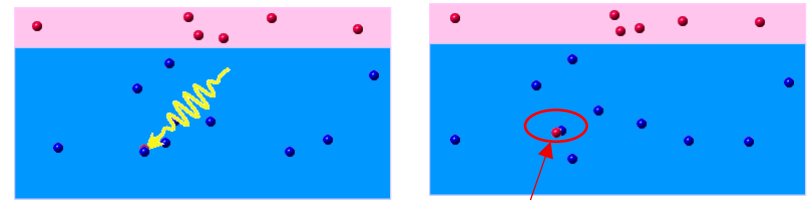
Photon generates an electron-hole pair in the base



Minority carrier (electron) is collected when it crosses the junction

Scenario 4

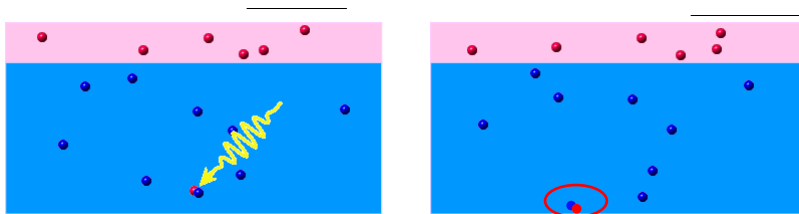
Photon generates an electron-hole pair near the rear surface



Minority carrier (electron) recombines in the base

Scenario 5

Photon generates an electron-hole pair close to the rear surface



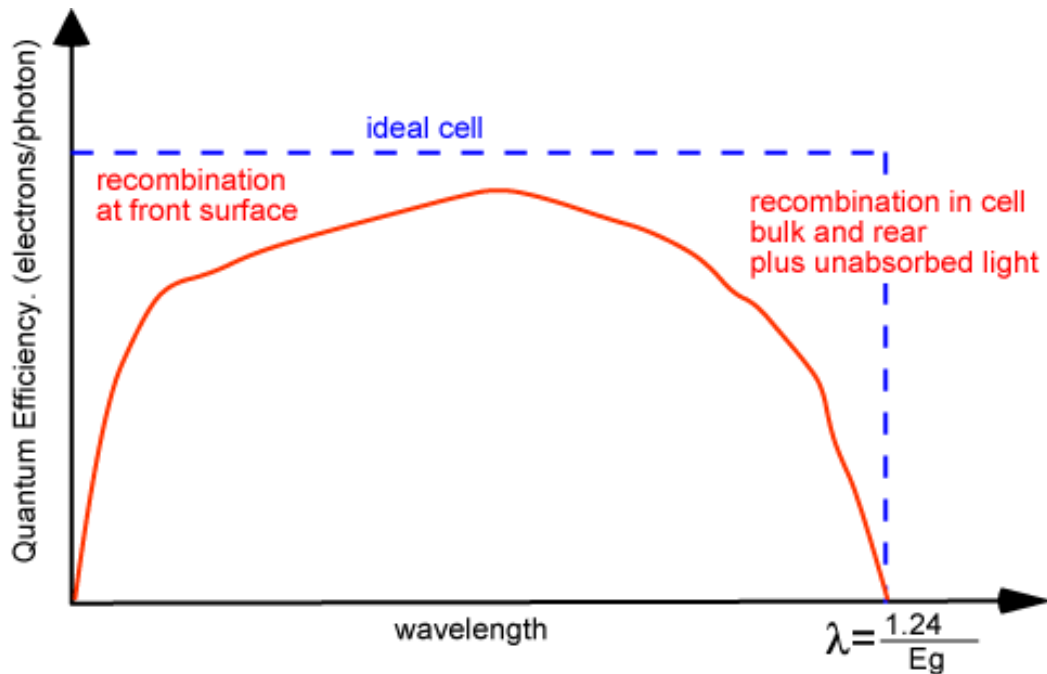
Minority carrier (electron) recombines at the rear surface

Current losses due to recombination

To reduce the current losses, both surface and bulk recombination must be minimized. In silicon solar cells, the two conditions commonly required for such current collection are:

1. the carrier must be generated within a diffusion length of the junction, so that it will be able to diffuse to the junction before recombining; and
2. in the case of a localized high recombination site (such as at an unpassivated surface or at a grain boundary in multicrystalline devices), the carrier must be generated closer to the junction than to the recombination site. For less severe localized recombination sites, (such as a passivated surface), carriers can be generated closer to the recombination site while still being able to diffuse to the junction and be collected without recombining.

The presence of localized recombination sites at the front and the rear surfaces reduces the collection probabilities of carriers generated from blue and infrared light. Since blue light has a high absorption coefficient and is absorbed very close to the front surface, it is not likely to generate minority carriers that can be collected by the junction if the front surface is a site of high recombination. Similarly, a high rear surface recombination will primarily affect carriers generated by infrared light, which can generate carriers deep in the device. The quantum efficiency of a solar cell quantifies the effect of recombination on the light generation current.

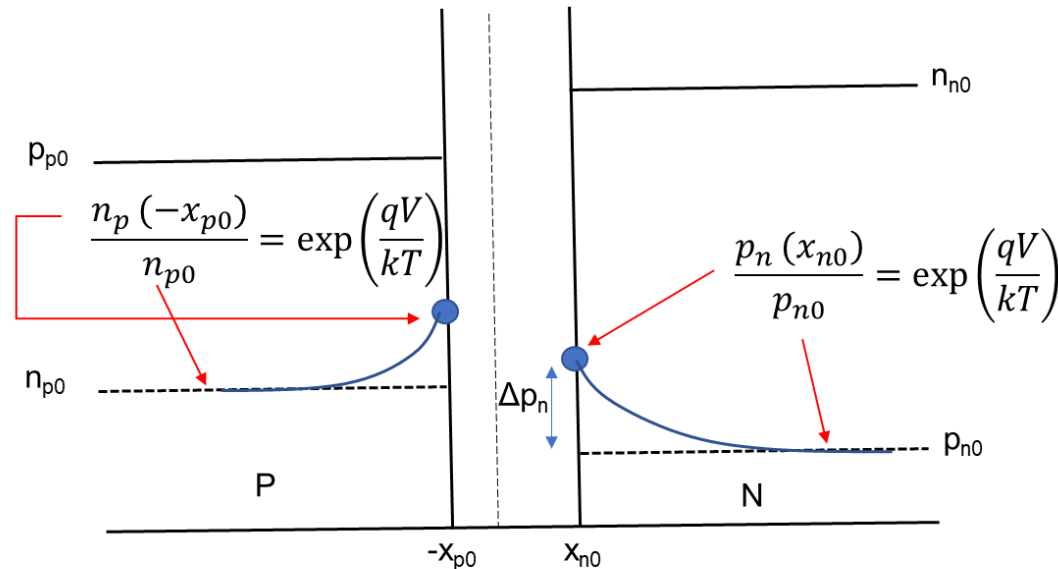


Typical quantum efficiency in an ideal and actual solar cell, illustrating the impact of optical and recombination losses.

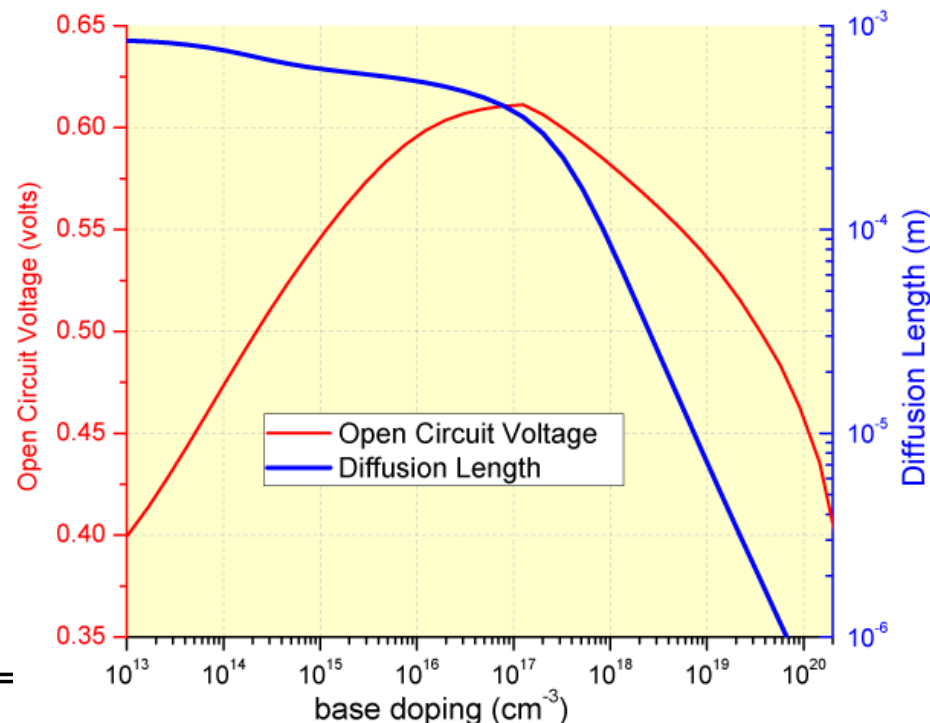
Voltage losses due to recombination

The open-circuit voltage is the voltage at which the forward bias diffusion current is exactly equal to the short-circuit current. The forward bias diffusion current depends on the amount recombination, which is controlled by the number of minority carriers at the junction edge, how fast they move away from the junction and how quickly they recombine. Consequently, the open-circuit voltage is affected by the following parameters:

- **the number of minority carriers at the junction edge:** The number of minority carriers injected from the other side is simply the number of minority carriers in equilibrium multiplied by an exponential factor which depends on the voltage and the temperature. Therefore, minimizing the equilibrium minority carrier concentration reduces recombination. Minimizing the equilibrium carrier concentration is achieved by **increasing the doping**;

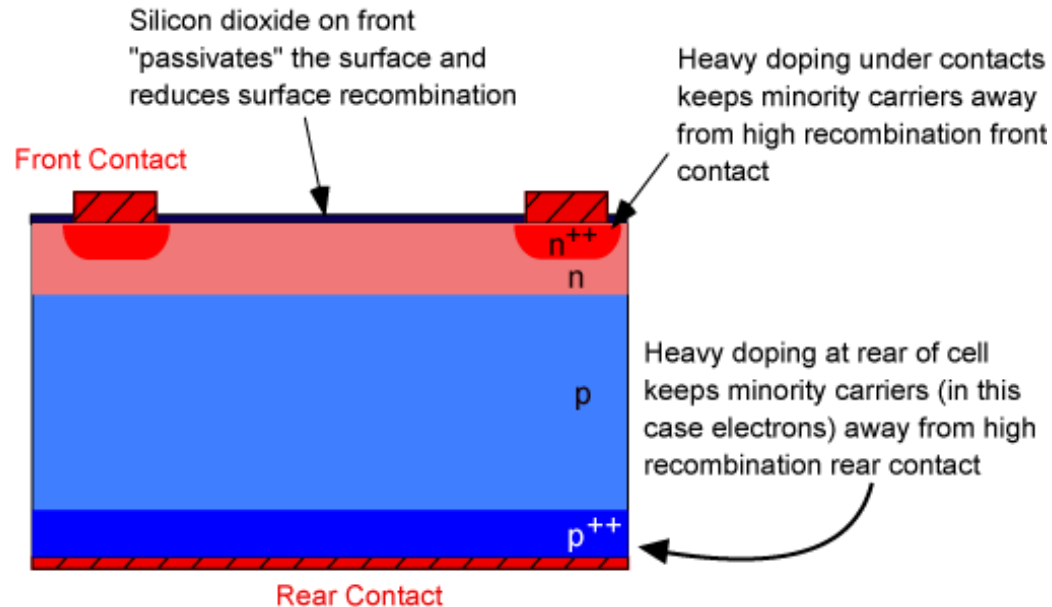


- **the diffusion length in the material:** To minimize recombination and achieve a high voltage, a longer diffusion length is required. The diffusion length depends on the types of material, the wafer processing and the doping in the wafer. High doping reduces the diffusion length, thus introducing a trade-off between maintaining a high diffusion length (which affects both the current and voltage) and achieving a high voltage;
- **the presence of localized recombination sources within a diffusion length of the junction:** A high recombination source close to the junction (usually a surface or a grain boundary) will allow carriers to move to this recombination source very quickly and recombine, thus dramatically increasing the recombination current and reduce the open circuit voltage.



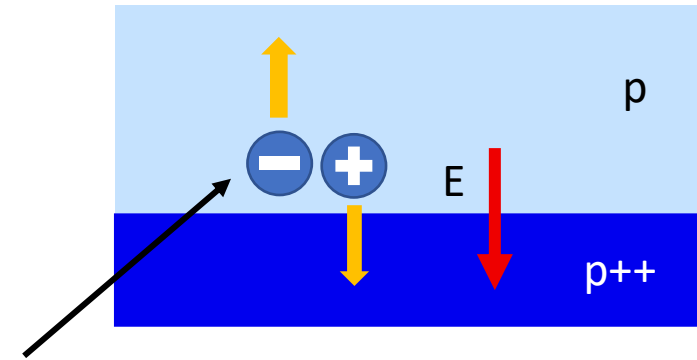
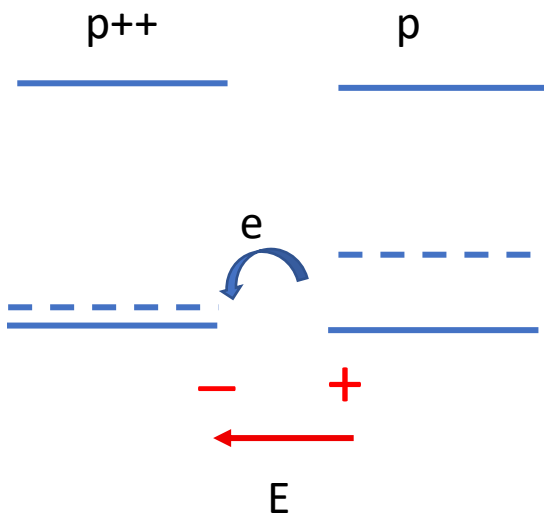
Surface Recombination

- Surface recombination can have a major impact both on the J_{sc} and V_{oc} .
- High recombination rates at the front surface have a particularly detrimental impact on the J_{sc} since the front surface also corresponds to the highest generation region of carriers in the solar cell.
- Lowering the high **front surface recombination** is typically accomplished by reducing the number of dangling silicon bonds at the front surface by using "**passivating**" layer on the front surface.
- It is possible to passivate the silicon surface with thin films of a variety materials, including **silicon dioxide, silicon nitride, aluminum oxide and amorphous silicon**.
- These thin films greatly reduce the density of surface defects while they are also serving as **anti-reflection coatings**.
- Additionally, the films may be electrically charged, which electrooptically repels either electrons or holes and hence reduces the recombination rate.

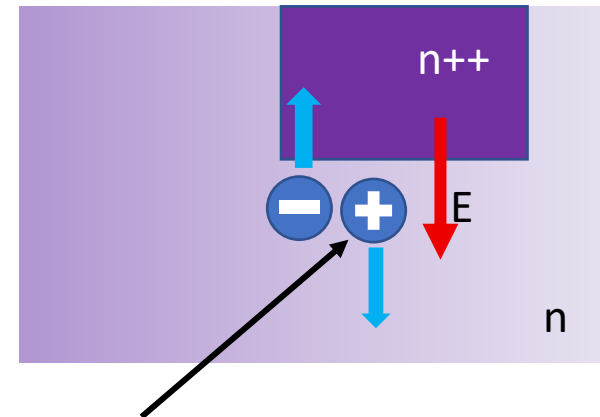
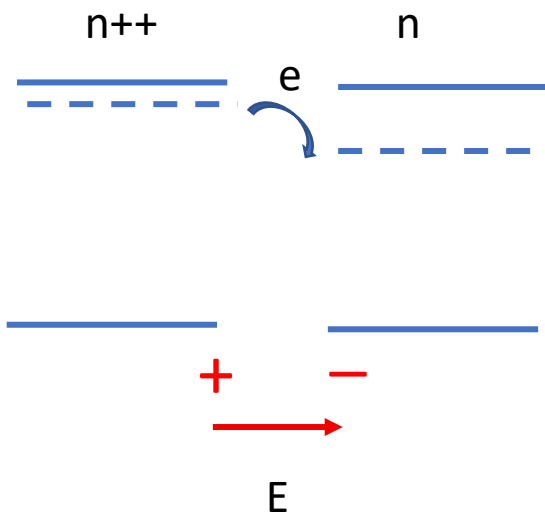


- Since the passivating layer for silicon solar cells is usually an insulator, any region which has an ohmic **metal contact cannot be passivated using silicon dioxide**.
- Instead, **under the top contacts the effect of the surface recombination can be minimized by increasing the doping**.
- While typically such a high doping severely degrades the diffusion length, the contact regions do not participate in carrier generation and hence the impact on carrier collection is unimportant.

- A similar effect is employed at the rear surface to minimize the impact of rear surface recombination velocity on voltage and current if the rear surface is closer than a diffusion length to the junction.
- A "back surface field" (BSF) consists of a higher doped region at the rear surface of the solar cell. The interface between the high and low doped region behaves like a p - n junction and an electric field forms at the interface which introduces a barrier to minority carrier flow to the rear surface. The minority carrier concentration is thus maintained at higher levels in the bulk of the device and the BSF has a net effect of passivating the rear surface.



Preventing the minority carrier coming to the back contact



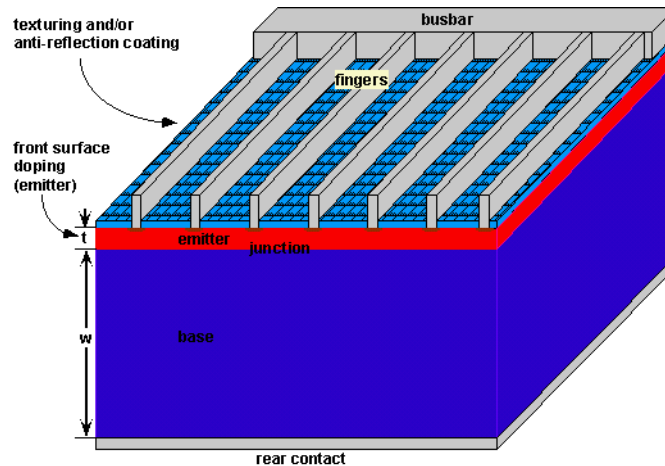
Preventing the minority carrier coming to the front contact

Bulk Recombination

- Within the bulk of silicon wafer common imperfections include foreign atoms, dislocations and other crystal defects, and grain boundaries, which are present in multicrystalline wafers.
- The use of high purity materials and clean process equipment reduces contamination.
- Some thermal steps remove impurities to reduce their concentration and activity.
- Crystal defects and grain boundaries can be minimized using specific thermal steps and passivation with hydrogen.
- There is an economic trade-off between the cost of minimizing crystal imperfections and the resulting increase in solar cell efficiency.

Structure of silicon solar cell

Basic design



The optimum thickness of cell with light trapping and surface passivation is about 100 μm but thickness between 200 and 500 μm are typically used, partly for practical issues such as making and handling thin wafers, and partly for surface passivation reasons. Basic cell design comprises of the following components.

Base (or) Substrate

Usually p-type substrate which is doped to reach the conductivity of $1 (\Omega\cdot\text{cm})^{-1}$. A higher base doping leads to a higher V_{oc} and lower resistance, but higher levels of doping result in damage to the crystal.

Emitter

N-type silicon has a higher surface quality than p-type silicon so it is placed at the front of the cell where most of the light is absorbed. Thus the top of the cell is the negative terminal and the rear of the cell is the positive terminal. A large fraction of light is absorbed close to the front surface. By making the front layer very thin ($<1\text{ }\mu\text{m}$), a large fraction of the carriers generated by the incoming light are created within a diffusion length of the p - n junction.

The front junction is doped to a level sufficient to conduct away the generated electricity without resistive losses. However, excessive levels of doping reduces the material's quality to the extent that carriers recombine before reaching the junction. Typically, emitter is doped to reach the sheet resistance of $100\text{ }\Omega/\text{sq}$.

The front surface of emitter is textured to increase the amount of light coupled into the cell.

Grid Pattern

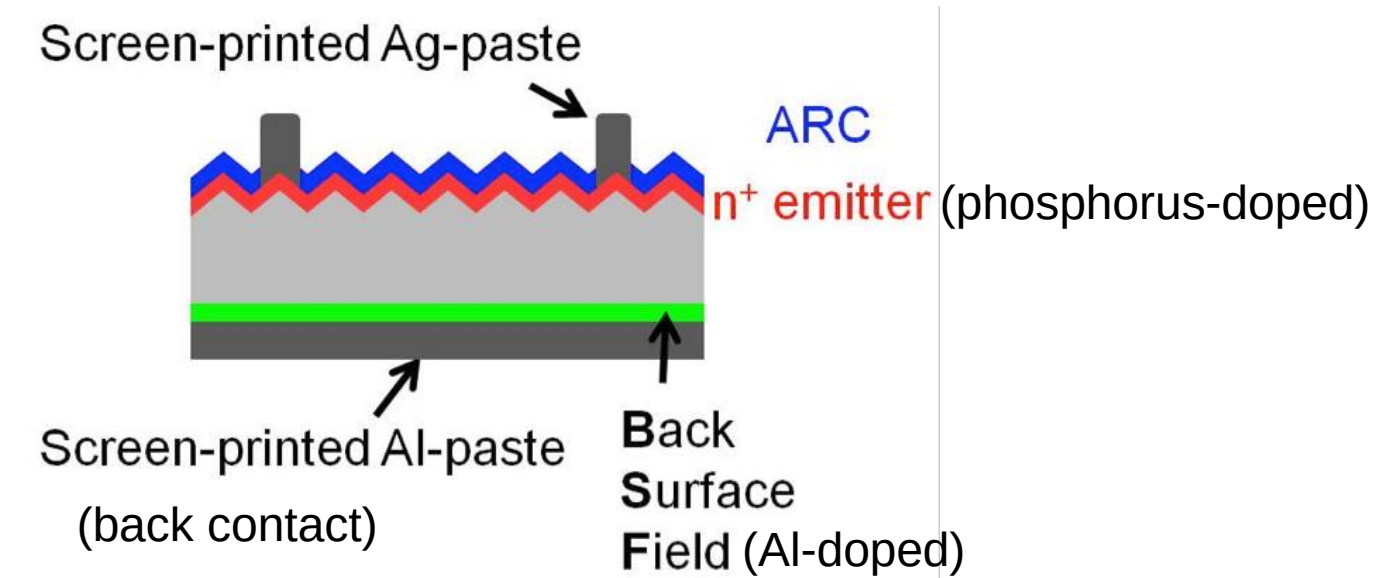
(fingers 20 to 200 μm width, placed 1 - 5 mm apart)

The resistivity of silicon is too high to conduct away all the current generated, so a lower resistivity metal grid is placed on the surface to conduct away the current. The metal grid shades the cell from the incoming light so there is a compromise between light collection and resistance of the metal grid.

Rear Contact

The rear contact is much less important than the front contact since it is much further away from the junction and does not need to be transparent. The design of the rear contact is becoming increasingly important as overall efficiency increases and the cells become thinner.

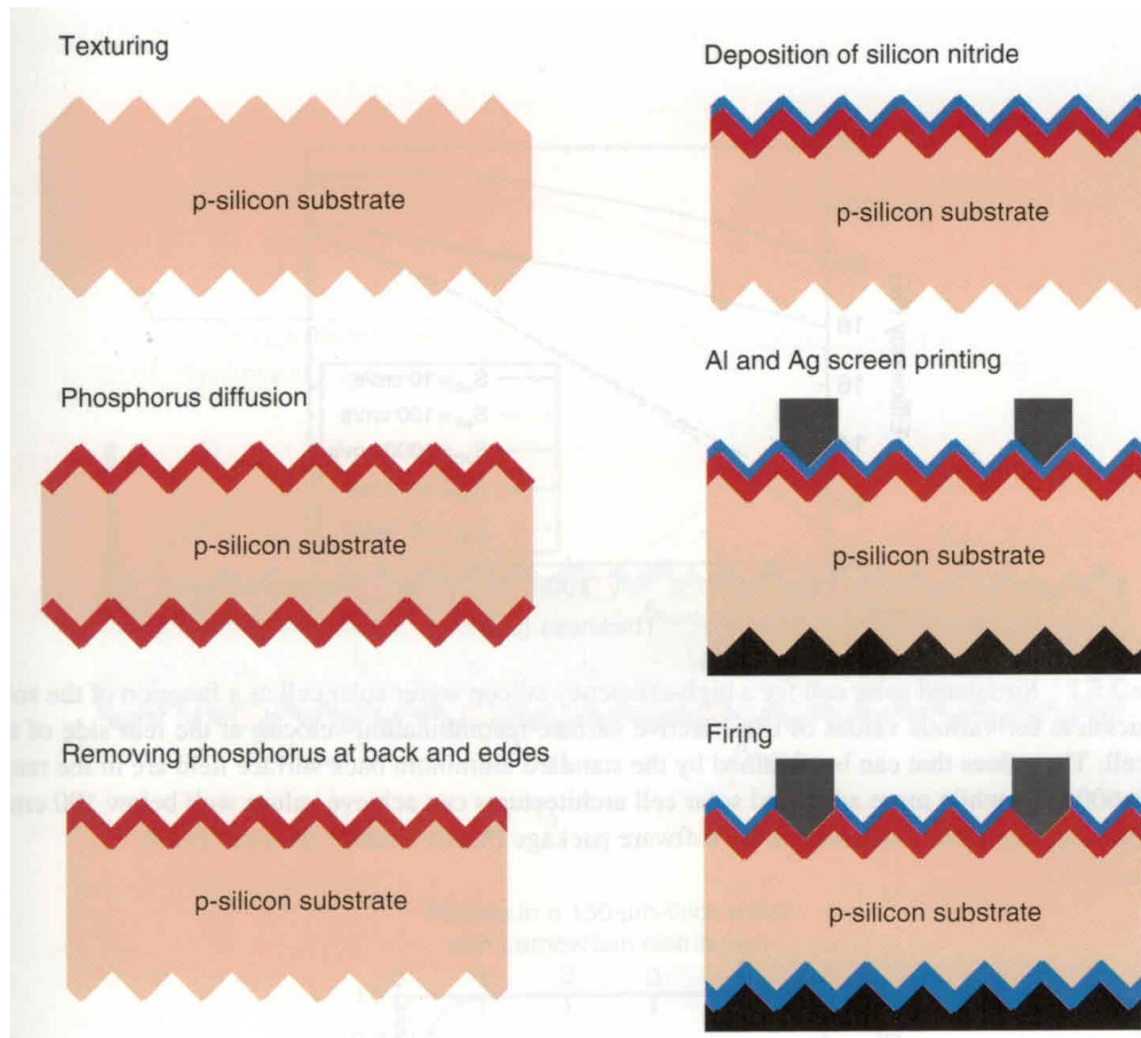
Screen Printed Silicon Solar Cell



- Leading commercial silicon solar cell technology
- Achieve efficiencies of 16-19% for monocrystalline silicon and 15-18% for multicrystalline silicon

Process steps

- Wafer texturing to reduce the reflection losses and improve light trapping
- Phosphorus doping to form n^+ emitter
- Removal of phosphorus doping at the edges and the back of wafer by plasma or chemical etching to avoid short-circuit between the front and rear surfaces
- Deposition of a thin silicon nitride layer on the front surface, which serves as a passivation as well as antireflection layer
- Screen printing of silver paste in a finger pattern on the front surface
- Screen printing of aluminum paste on the rear surface (Note: pastes consists of small metal particles together with solvents and binders)
- Firing the paste in a furnace to bind the metal particles together and form electrically conductive layers, and to diffuse metal fingers through silicon nitride (ARC) layer.
- On the rear surface, the fired aluminum paste causes a back surface field (BSF) to form, which reduces recombination rates at the rear surface.

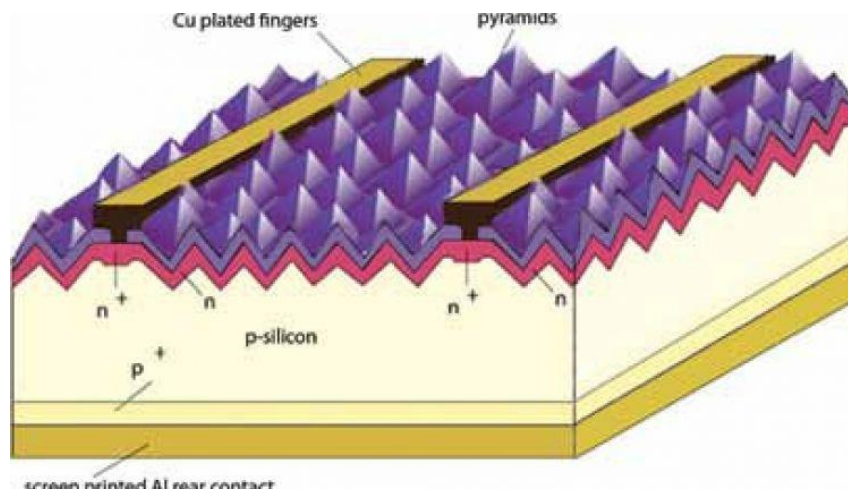


A typical process of a screen-printed silicon cell

Selective Emitter Solar Cell

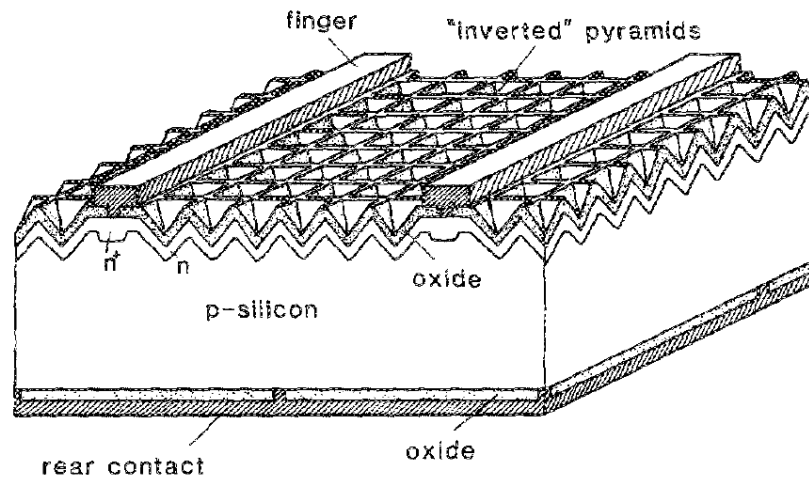
- The contact resistance at the finger electrodes is usually high due to a narrow width of fingers to reduce the optical losses.
- The contact resistance between the finger electrodes and the silicon can be reduced if the surface of the silicon is relatively heavily doped with phosphorus.
- In addition, the lateral resistance within the phosphorous-doped layer is reduced by the heavier doping.
- However, heavy phosphorous doping near the surface leads to reduced responsiveness to the blue and UV light, which reduces the cell efficiency.
- In selective emitter solar cell, phosphorous is heavily doped beneath the finger electrodes only and relatively light doped over the rest of the surface.
- The blue response of the cell is improved while maintain low contact resistance and low recombination at the metal contact.
- Additional process steps -> higher cost

Selective Emitter Solar Cell



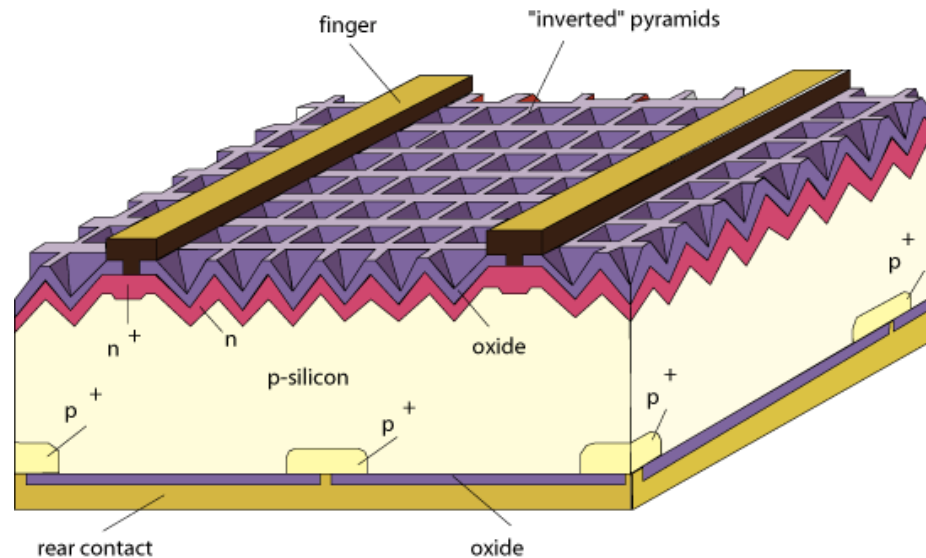
PERC (Passivated Emitter and Rear Cells)

- The screen printed aluminum layer has relatively high parasitic absorbance of light and relatively high recombination rate.
- A thin dielectric film is placed between the silicon and the rear metal contact to substantially reduce both optical absorption losses and recombination.
- An array of holes in the dielectric film (covering about 1% of the rear surface and spacing about a millimetre apart) allows the electrical contact to be made directly to the rear surface of the p-silicon wafer.
- The improved performance of a PERC cell has to be balanced against the slightly higher cost of cell fabrication.



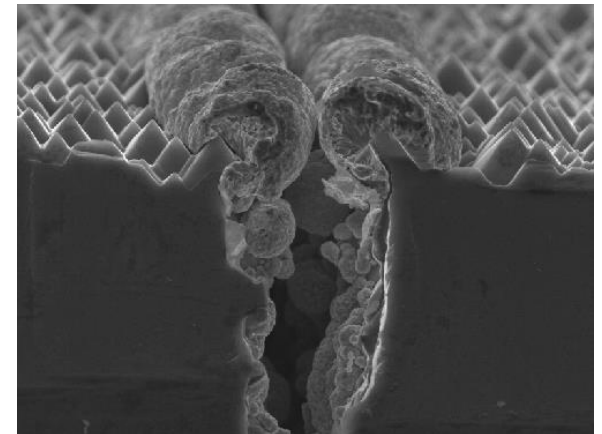
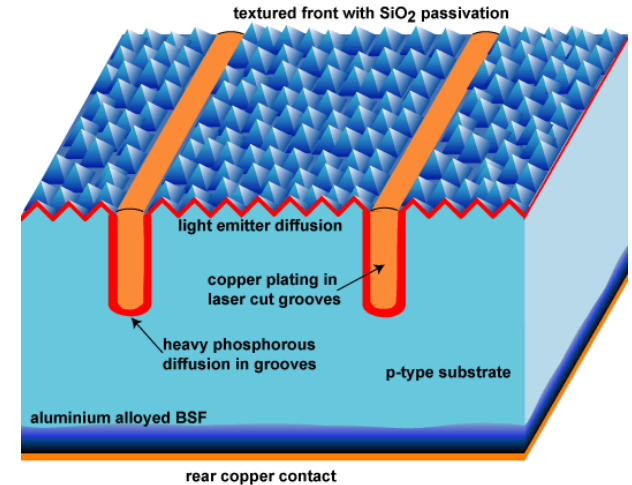
PERL (Passivated Emitter and Rear Local Diffusion)

- Similar design to PERC solar cells with the addition of boron diffusions at the rear contact points to reduce recombination and resistive losses.
- Efficiency of 25%
- The manufacturing process is complicated and have not yet entered commercial production.



Buried contact solar cell

- technology based on a plated metal contact inside a laser-formed groove
- 25% better than commercial screen-printed solar cells.
- A large metal height-to-width aspect ratio allows a large number of closely spaced metal fingers, while still retaining a high transparency.
- Low parasitic resistance losses due to high metal aspect ratio
- The metal grid resistance is also low due to the large volume of metal in the grooves and by the use of copper.
- Low contact resistance due to the large metal-silicon contact area

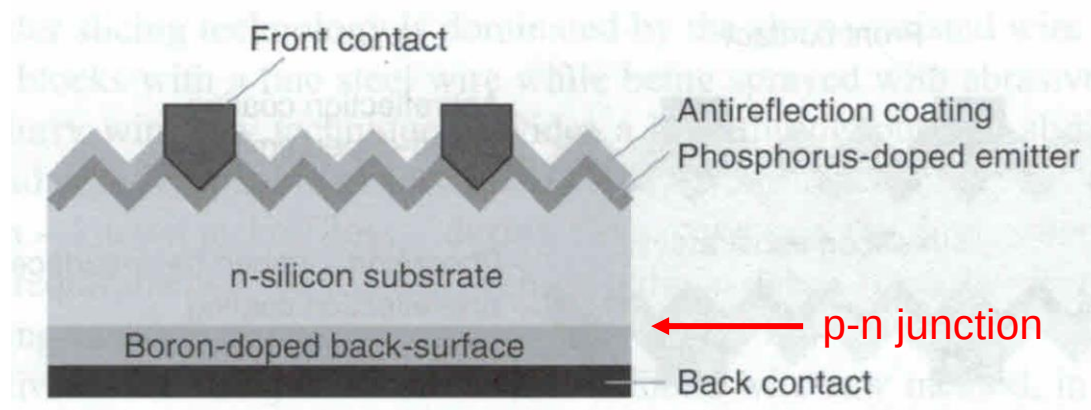


Switching to n-type substrate

- The photovoltaic industry has been dominated by boron-doped p-type silicon
- p-type silicon has boron-oxygen defects and greater recombination strength.
- Therefore, p-type silicon suffers from reduced minority carrier lifetime as compared to n-type silicon
- The solar cells fabricated on phosphorous-doped silicon (n-type) has performance advantage.
- Interdigitated back contact solar cells and heterojunction back contact silicon solar cells typically use n-type substrate.

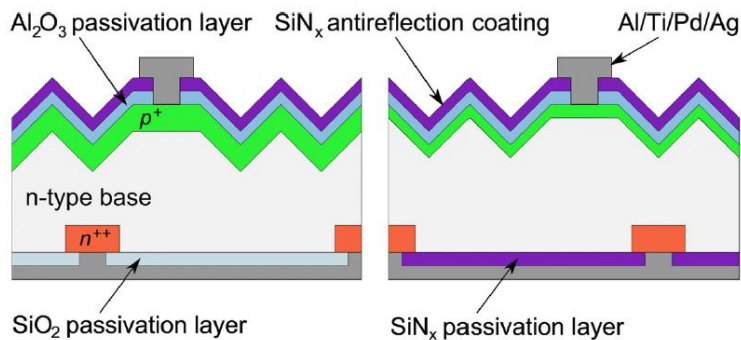
N-type rear emitter silicon solar cell

- Substrate – phosphorous-doped (n-type) silicon wafer
- The p-n junction is located at the rear surface
- The p-type emitter (p+) is formed by aluminum-alloying the silicon or by boron diffusion.
- The photo-generated carriers have to diffuse to the rear surface to be collected by the junction
- Thus, higher quality substrate as well as effective front passivation and thinner wafer are required.

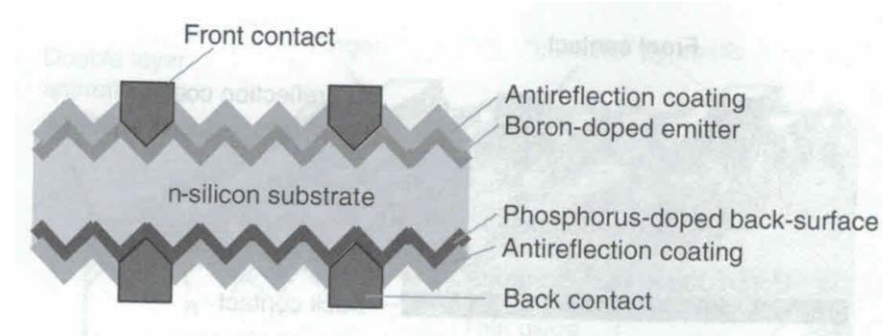


N-type front emitter silicon solar cell

- High-quality n-type silicon substrate is not required because photogenerated carriers can be collected at the front emitter.

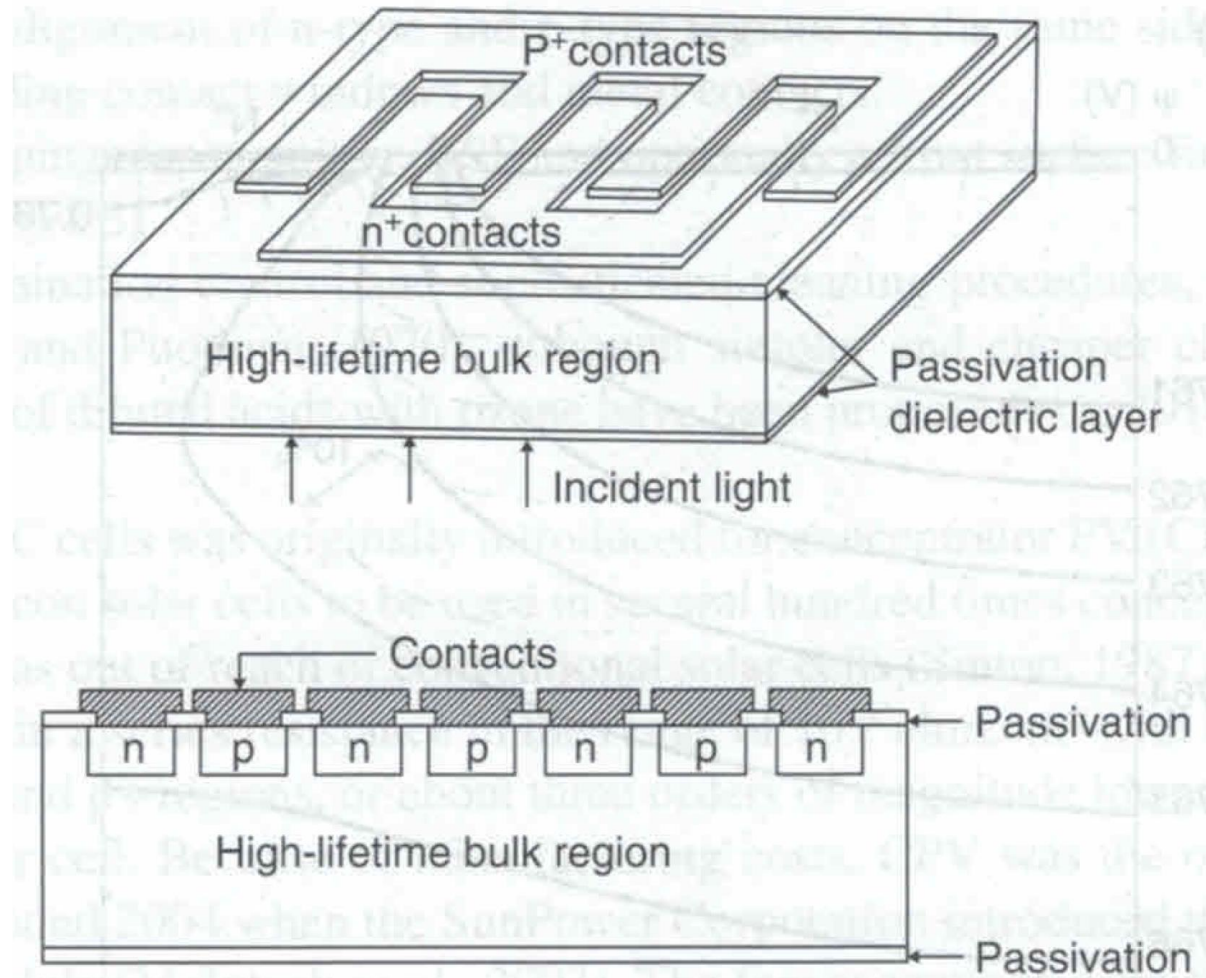


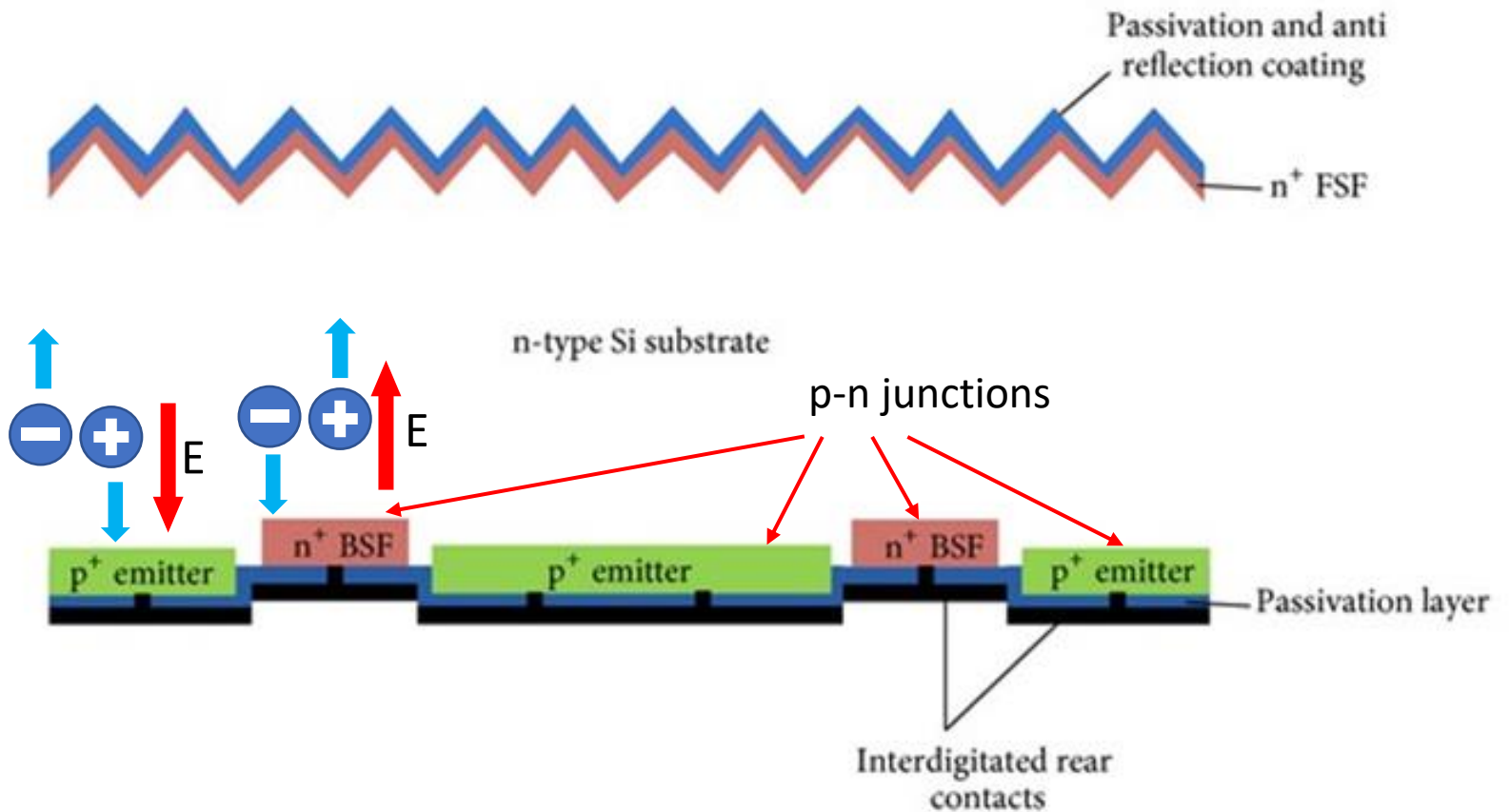
The p^+ emitter at the front and n^+ diffusion at the rear (p^+nn^+ structure, or PERL structure on n-type substrate)



The bifacial solar cell on n-type substrate; benefit of collecting light from both sides of the cell.

Interdigitated Back Contact Solar Cells





Interdigitated Back Contact Solar Cells

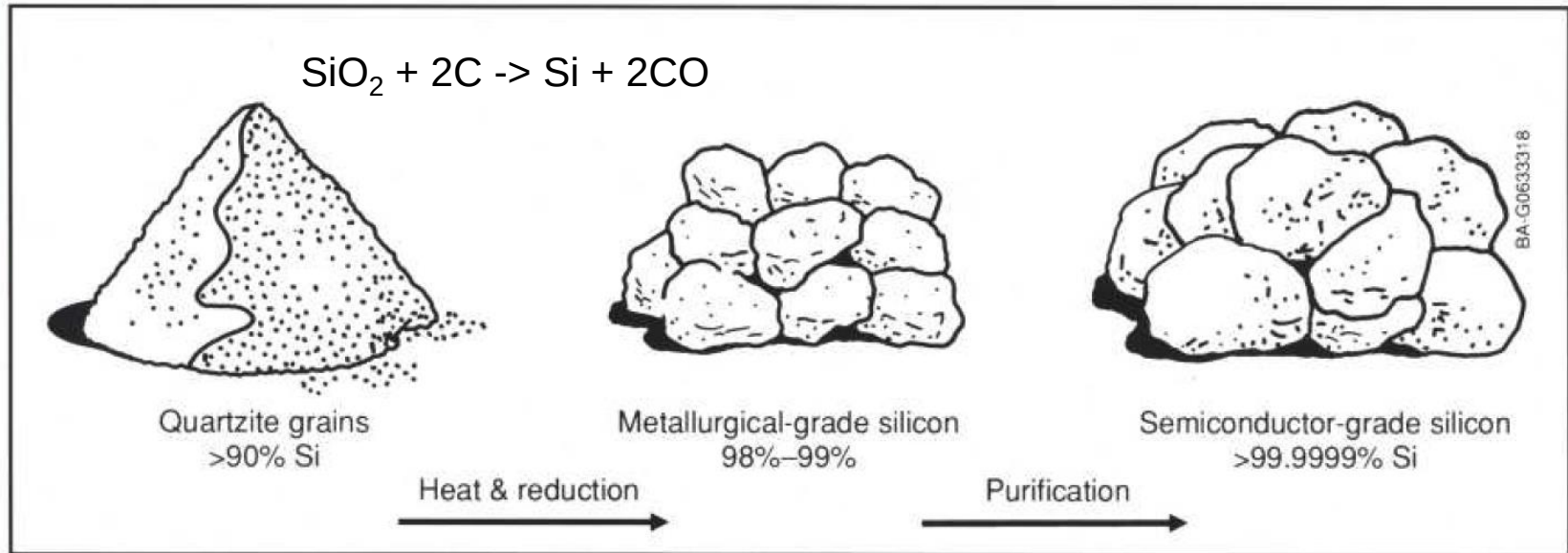
- IBC cells can be understood as rear junction solar cell.
- In a rear junction cell, the photo-generated carriers are generated close the front surface and have to travel all the way through the thickness of the silicon wafer, with the minority carries toward the emitter and the majority carriers toward the bulk contact or back surface field (BSF)
- The efficiency of the IBC cell is, therefore, very sensitive to the bulk lifetime and the front surface recombination.
- It requires a long minority carrier lifetime and a low front surface recombination velocity to achieve a decent efficiency.
- IBC design requires a minority carrier diffusion length much greater (about 5 to 10 times) than the thickness of the solar cell, equivalent to a carrier lifetime greater than 1 ms and a front surface recombination parameter J_0 smaller than 10fA/cm^2 .
- As a result of these requirements, the fabrication process is complicated and more expensive than that of conventional cells.

Interdigitated Back Contact Solar Cells

- No shadow loss from the front electrode, corresponding to a gain of about 5-7% in photogenerated current compared to conventional solar cells
- Low series resistance because each electrode can cover almost 50% of the back surface of the solar cell
- Decoupling of the optical optimization of the front side and the electrical optimization of the front side of the cell
- The efficiency is not limited by the trade-off between the sheet resistance of a front emitter and the recombination parameter J_0 of the same front emitter
- The front surface, where the largest concentration of minority carriers is located, can be optimized to obtain a very low surface recombination velocity S or a low recombination parameter J_0
- Easy to assemble into modules with ribbons or interconnects since both polarities are on the same side of the cell.

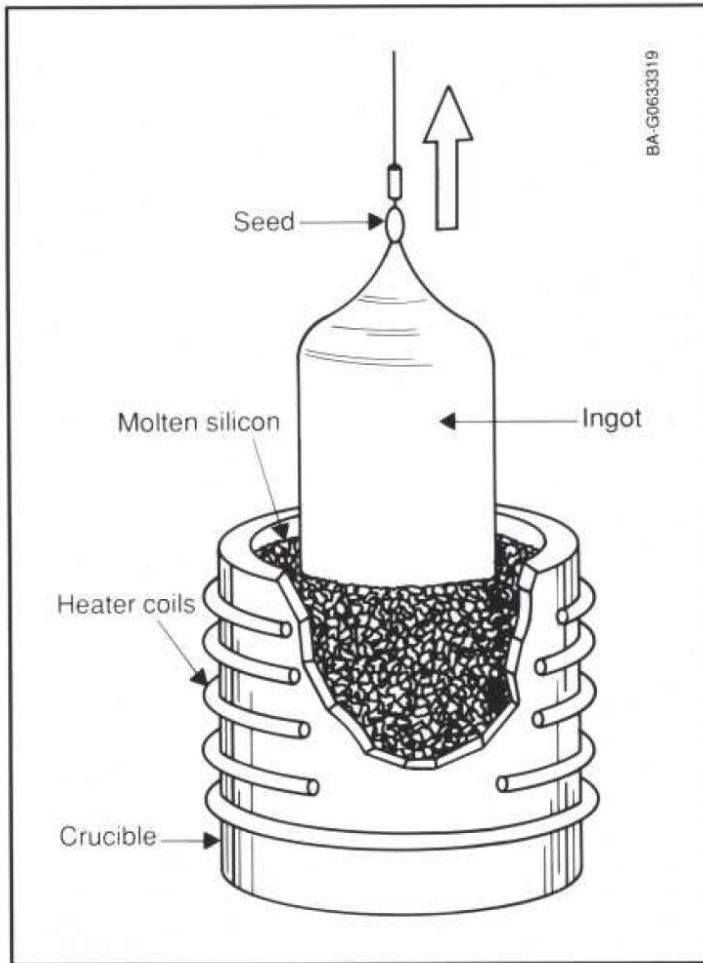
Manufacturing Silicon Solar Cell

Processing Silicon



To get the pure silicon required for solar cells, grains of quartzite, which are 90% silicon, are heated and reduced to produce metallurgical-grade silicon, which is 98% to 99% pure. This material is then refined to produce semiconductor-grade silicon, a product that is at least 99.9999% pure silicon.

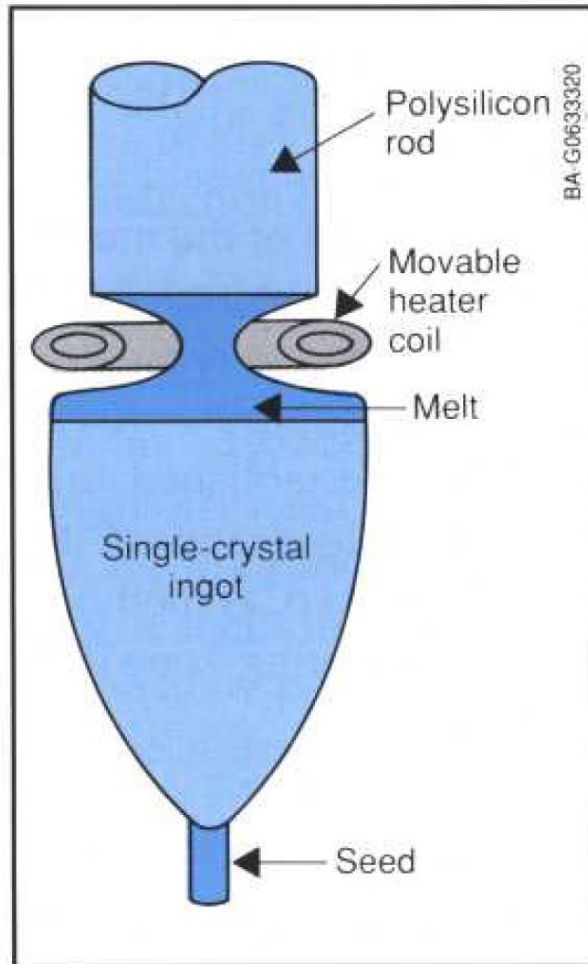
Preparing Single Crystal Silicon



Czochralski (CZ) method

The most widely used technique for making single-crystal silicon is the Czochralski process, in which a seed of single-crystal silicon contacts the top of molten silicon. As the seed is slowly raised, atoms of the silicon melt solidify in the pattern of the seed and extend the single-crystal structure.

Preparing Single Crystal Silicon

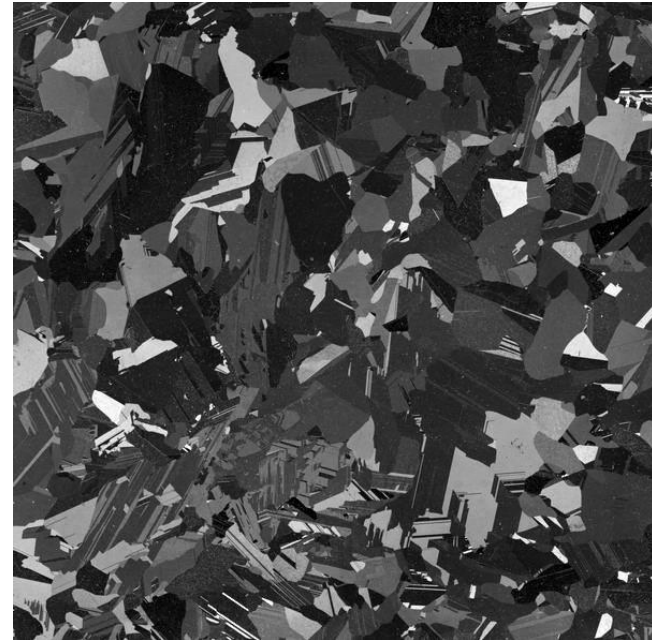


Float-zone (FZ) method

In the floating-zone technique, a silicon rod is set atop a seed crystal. Movable heating coils melt the interface between the seed and the silicon. Single-crystal material solidifies at the interface and moves upward as the heating coils are slowly raised.

Preparing multicrystalline silicon

The ingot growth for multicrystalline silicon is quite simple, melt the silicon in a large crucible and let it cool slowly to form large crystal. The specifics of furnace design allows the ingot to cool slowly so that very large grains (> 1 cm) are formed.



Other processing steps

- Sawing the ingot into bricks
- Wafer Slicing
- Texturing
- Emitter Diffusion
- Edge Isolation
- Anti Reflection Coatings
- Screen Print Front
- Screen Print Rear
- Firing
- Testing